

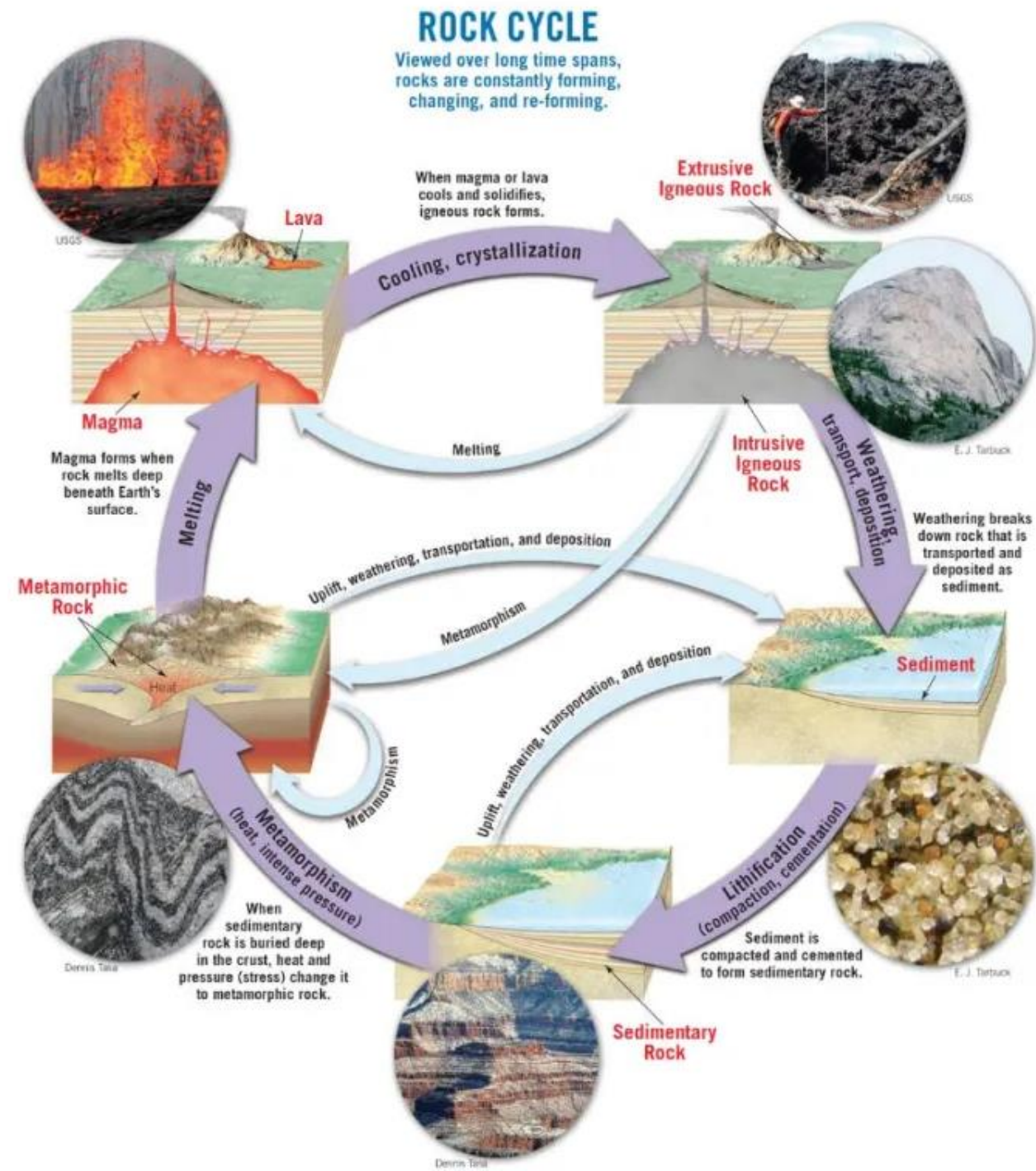
Class 4: Metamorphic Rocks

GS 315

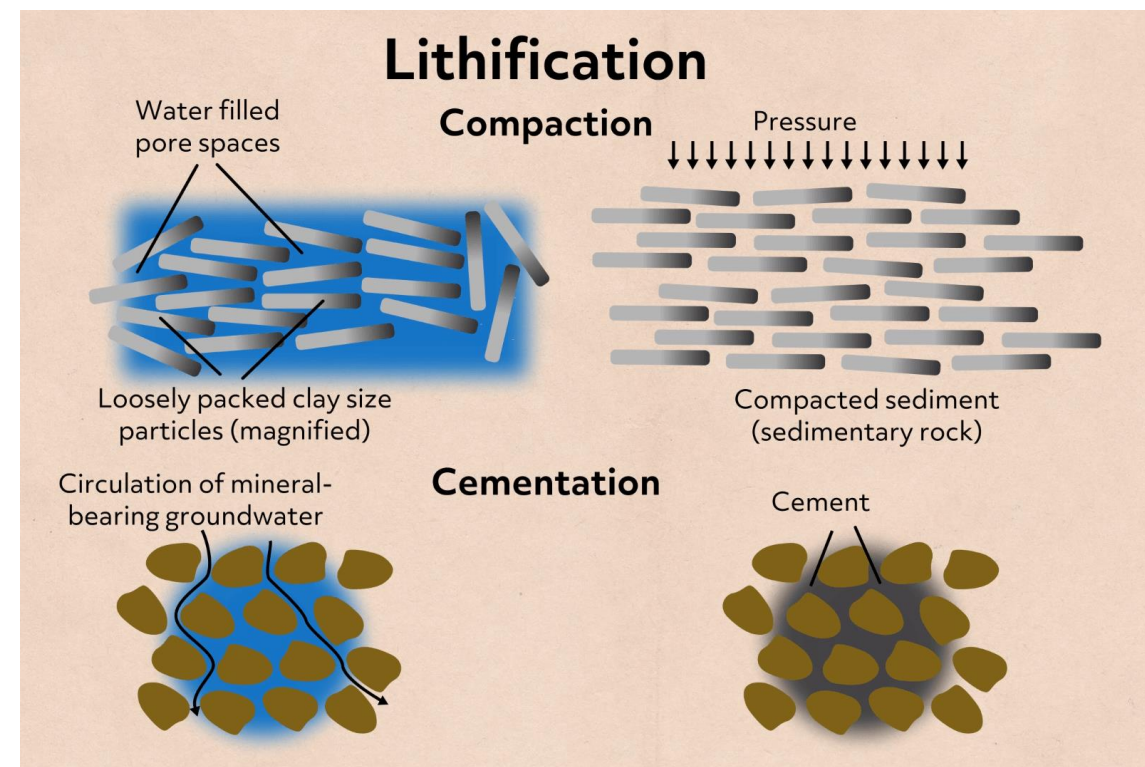
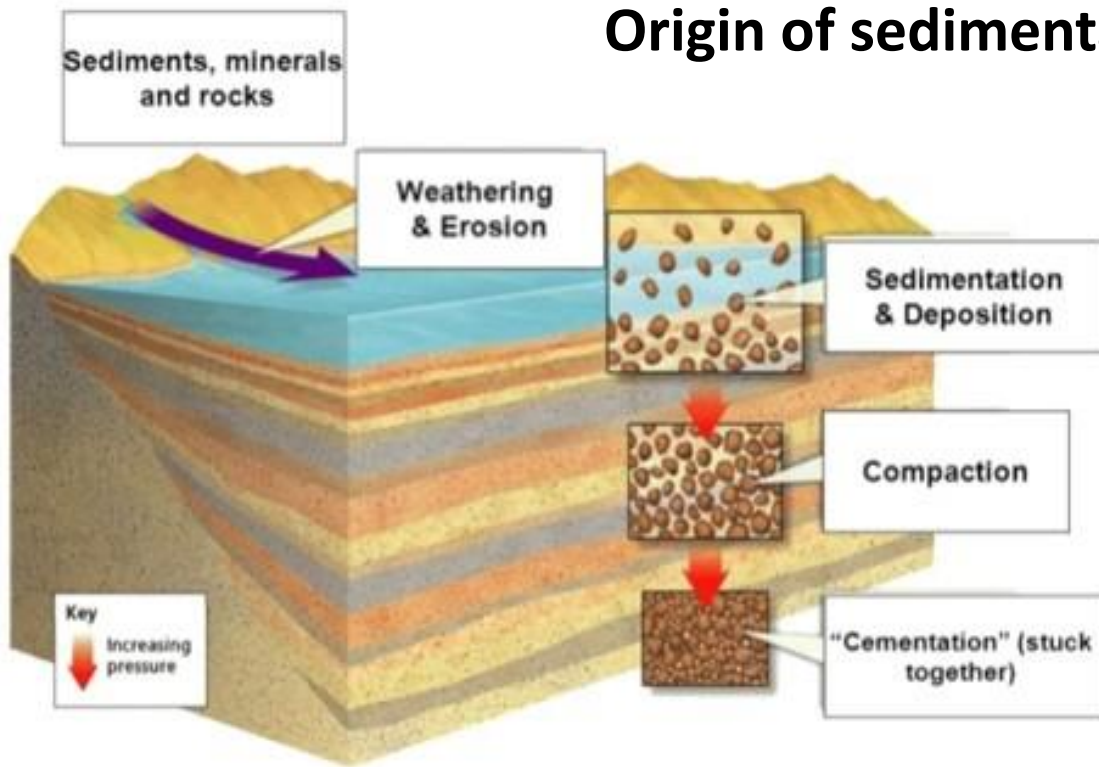


Instructor: Nibedita Rakshit

- **Igneous rocks** are formed from magma, which has originated well below the surface, has ascended towards the surface, and has crystallised as solid rock either on the surface or deep within the Earth's crust as its temperature fell.
- **Sedimentary rocks** are formed by the accumulation and compaction of (a) fragments from pre-existing rocks which have been disintegrated by erosion; (b) organic debris such as shell fragments or dead plants; or (c) material dissolved in surface waters (rivers, oceans, etc.) or ground water, which is precipitated in conditions of oversaturation.
- **Metamorphic rocks** are formed from pre-existing rocks of any type, which have been subjected to increases of temperature (T) or pressure (P) or both, such that the rocks undergo change. This change results in the metamorphic rock being different from the original parental material in appearance, texture and mineral composition.



Origin of sedimentary Rocks



- Weathering begins the process.
- Next, gravity and erosional agents (running water, wind, waves, and glacial ice) remove the products of weathering and carry them to a new location, where they are deposited.
- Usually, the particles are broken down further during this transport phase.
- Following deposition, this sediment may become lithified, or "turned to rock."
- Commonly, compaction and cementation transform the sediment into solid sedimentary rock.
- Weathered debris is constantly being swept from bedrock and carried away by water, ice, or wind.
- Eventually, the material is deposited in lakes, river valleys, seas, and countless other places.
- As piles of sediment accumulate, the materials near the bottom are compacted by the weight of the overlying layers.
- Over long periods, these sediments are cemented together by mineral matter deposited from water in the spaces between particles. This forms solid sedimentary rock.

Clastic/Detrital
Sedimentary
rocks

Detrital Sedimentary Rocks

Clastic Texture (particle size)	Sediment Name	Rock Name
Coarse (over 2 mm)	Gravel (Rounded particles)	Conglomerate 
	Gravel (Angular particles)	Breccia 
Medium (0.063 – 2 mm)	Sand	Sandstone 
		Arkose* 
Fine (0.0039 - 0.063 mm)	Silt	Siltstone 
Very fine < 0.0039 mm)	Clay	Shale or Mudstone 

*If abundant feldspar is present the rock is called arkose.

Detrital sedimentary rocks are classified by particle size, in certain cases, the mineral composition is also part of naming a rock

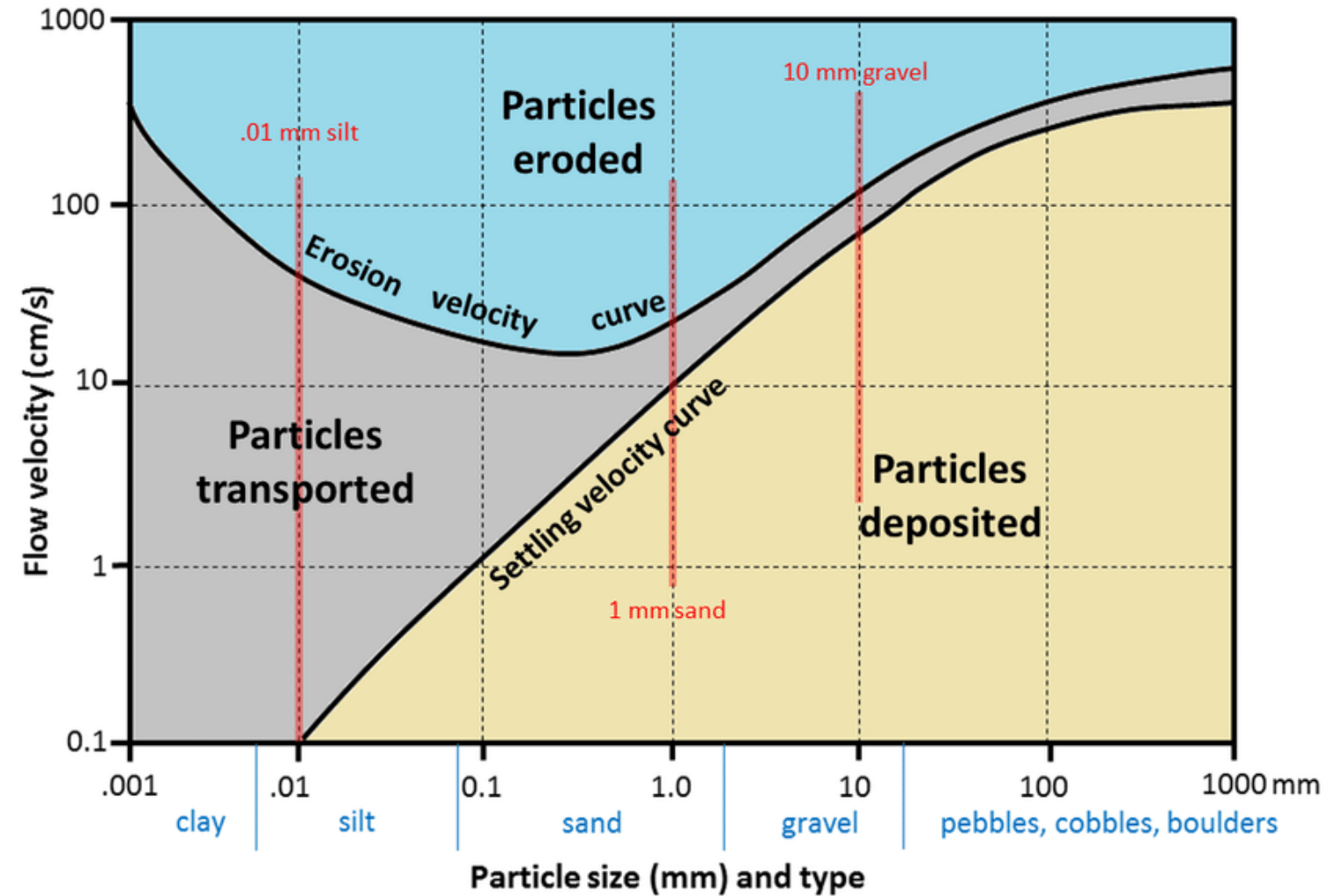
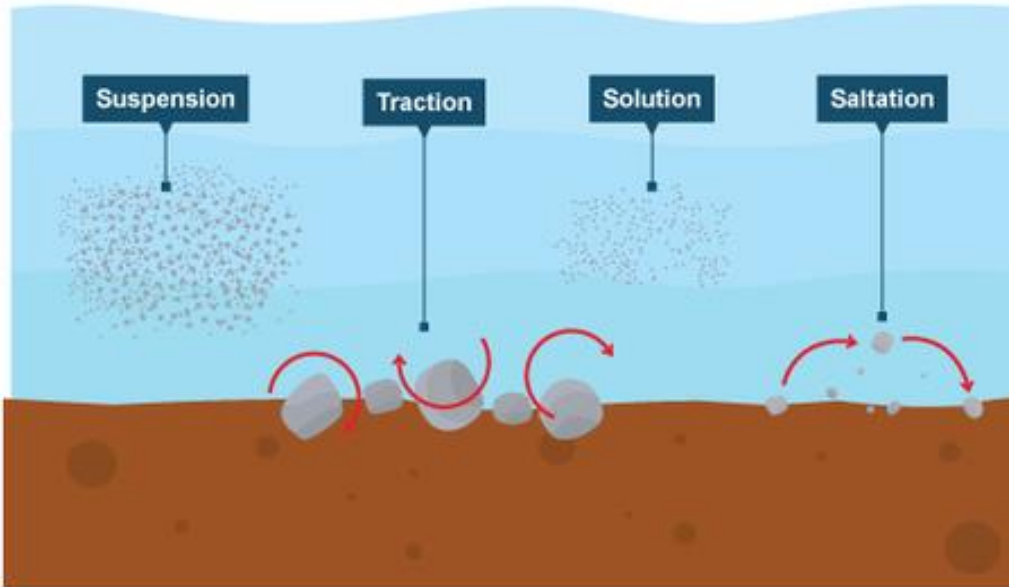
Gravel-size particles predominate, the rock is called conglomerate if the sediment is rounded and breccia if the pieces are angular. Angular fragments indicate that the particles were not transported very far from their source prior to deposition and so have not had corners and rough edges abraded.

Sandstone is the name given to rocks when sand-size grains prevail.

Siltstone, another rather fine-grained rock, is composed of clay-sized sediment intermixed with slightly larger silt sized grains.

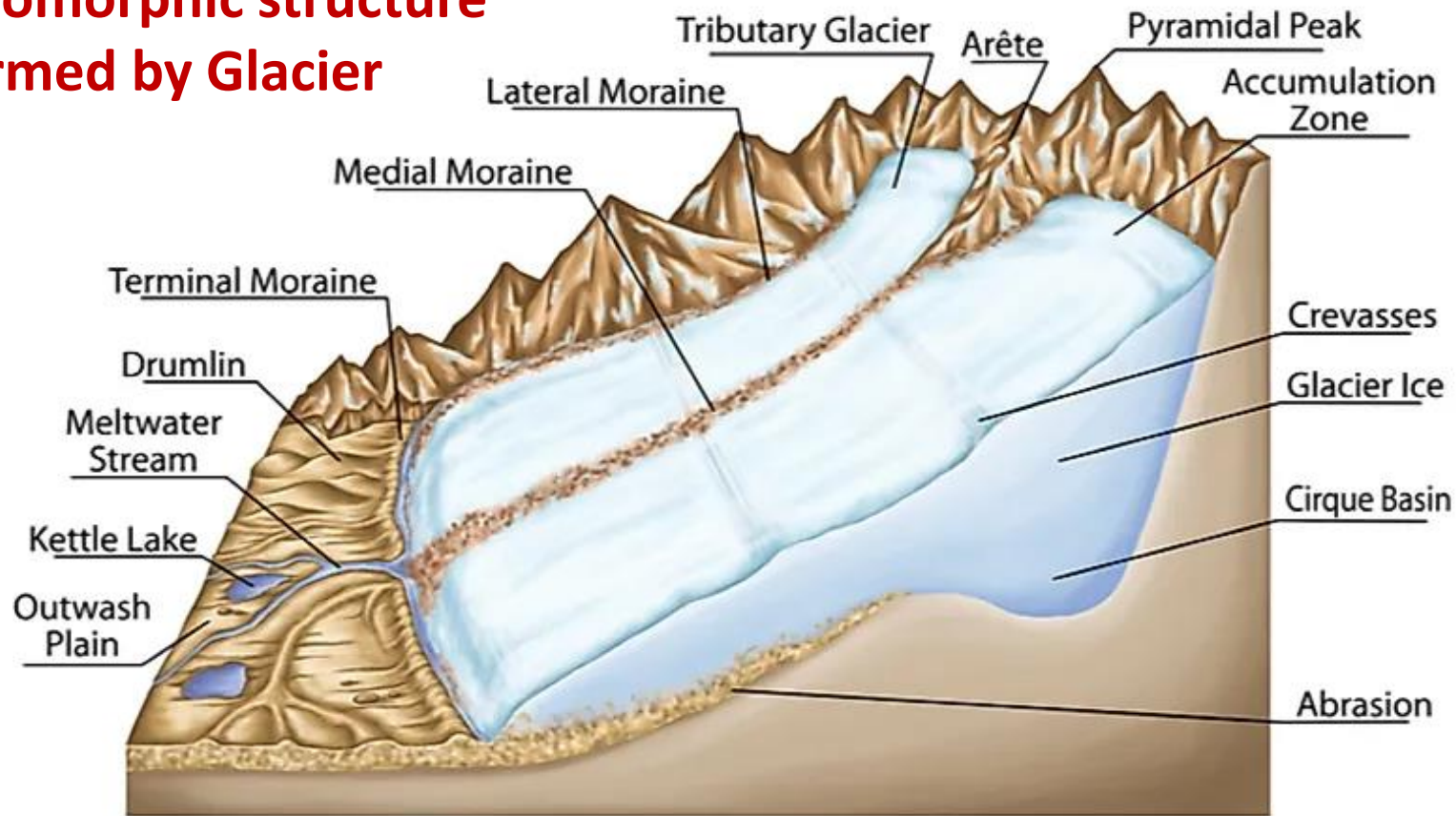
Shale, the most common sedimentary rock, is made of very fine-grained sediment and composed mainly of clay minerals.

Hjülstrom-Sundborg Diagram



- Currents of water or air sort the particles by size.
- The stronger the current, the larger the particle size carried.
- Gravels, for example, are moved by swiftly flowing rivers, rockslides, and glaciers.
- Less energy is required to transport sand; thus, it is common in windblown dunes, river deposits, and beaches.
- Because silts and clays settle very slowly, accumulations of these materials are generally associated with the quiet waters of a lake, lagoon, swamp, or marine environment.

Geomorphic structure formed by Glacier

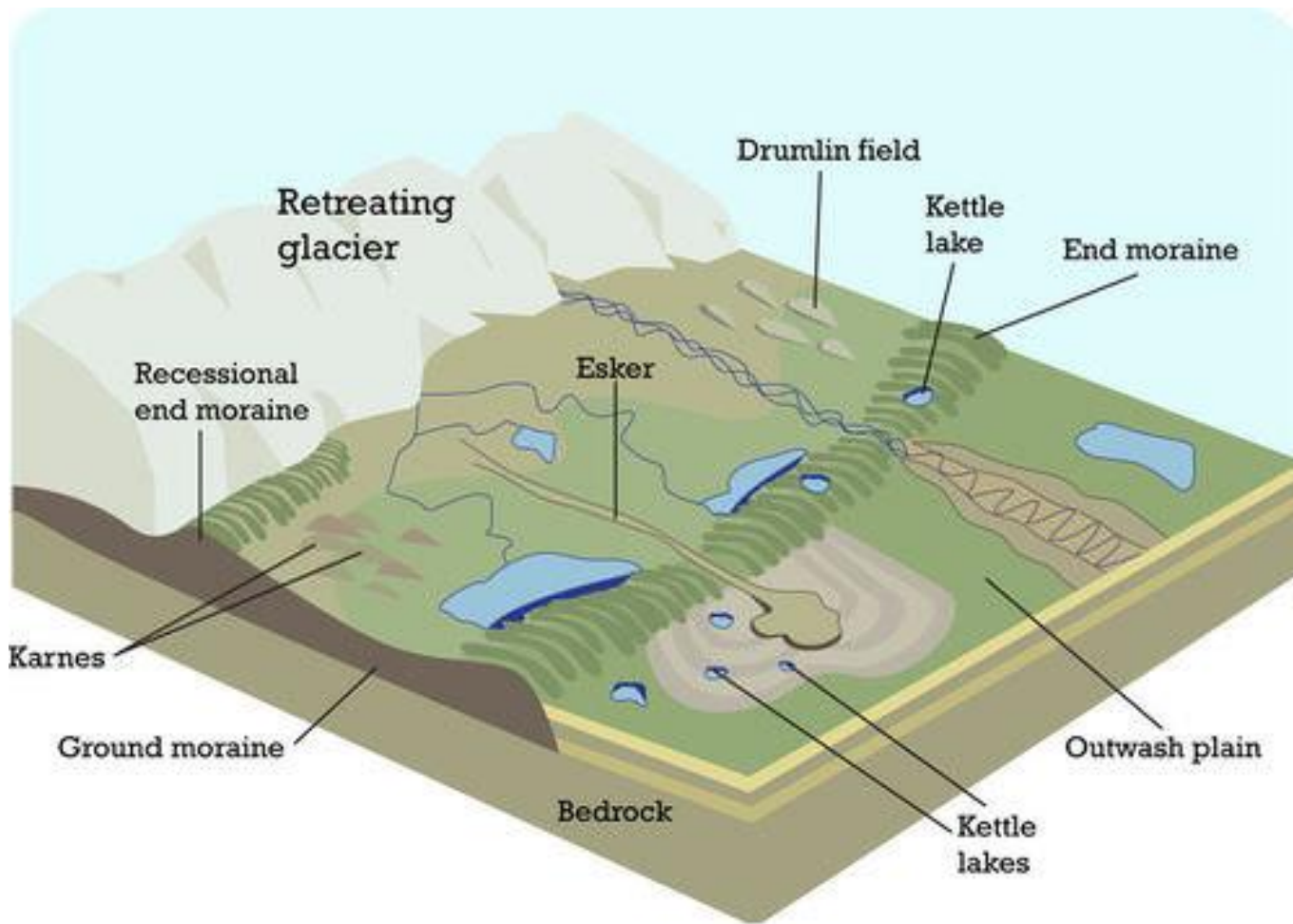


Erosional landforms

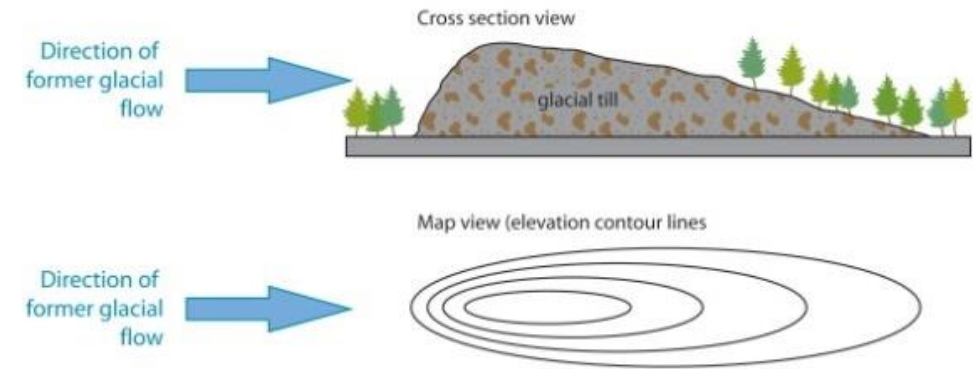
- Cirque: Starting location for mountain glaciers
- Cirque stairway: a sequence of cirques
- U-shaped, or trough, valley: U-shaped valleys are created by mountain glaciers. When filled with ocean water so as to create an inlet, these valleys are called fjords.
- Arête: spiky high land between two glaciers, if the glacial action erodes through, a spillway (or col) forms
- Valley step: an abrupt change in the longitudinal slope of a glacial valley

Depositional landforms

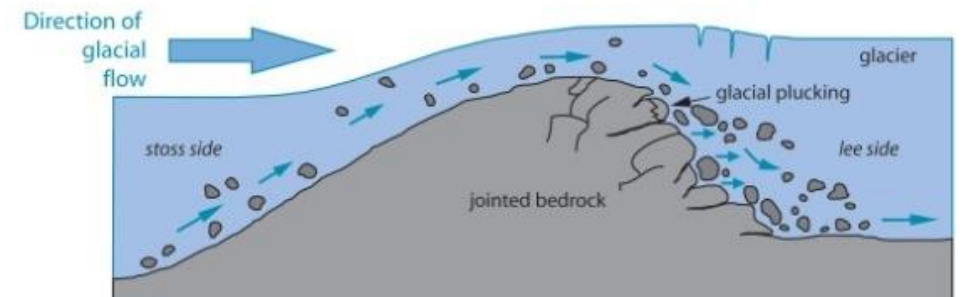
- Moraines are also landforms left behind by retreating glaciers. Many depositional landforms result from sediment deposited or reshaped by meltwater and are referred to as fluvioglacial landforms.
- Esker: Built up bed of a subglacial stream.
- Kame: Irregularly shaped mound.
- Moraine: Feature can be terminal (at the end of a glacier), lateral (along the sides of a glacier), or medial (formed by the merger of lateral moraines from contributory glaciers).
- Outwash fan: Braided stream flowing from the front end of a glacier.



Drumlins



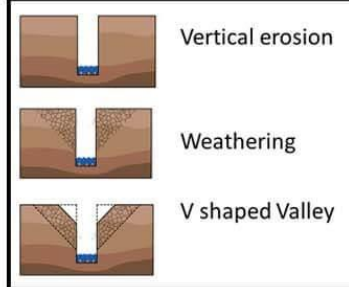
Roche moutonnées



- A **drumlin**, from the Irish word droimnín ("little ridge"), in the classical sense is an elongated hill in the shape of an inverted spoon or half-buried egg formed by glacial ice acting on underlying unconsolidated till or ground moraine.
- Assemblages of drumlins are referred to as fields or swarms.
- They can create a landscape which is often described as having a 'basket of eggs topography'.

Upper Course

flows with high velocity
V-shaped valley
vertical erosion dominates
waterfalls, rapids, gorges etc.

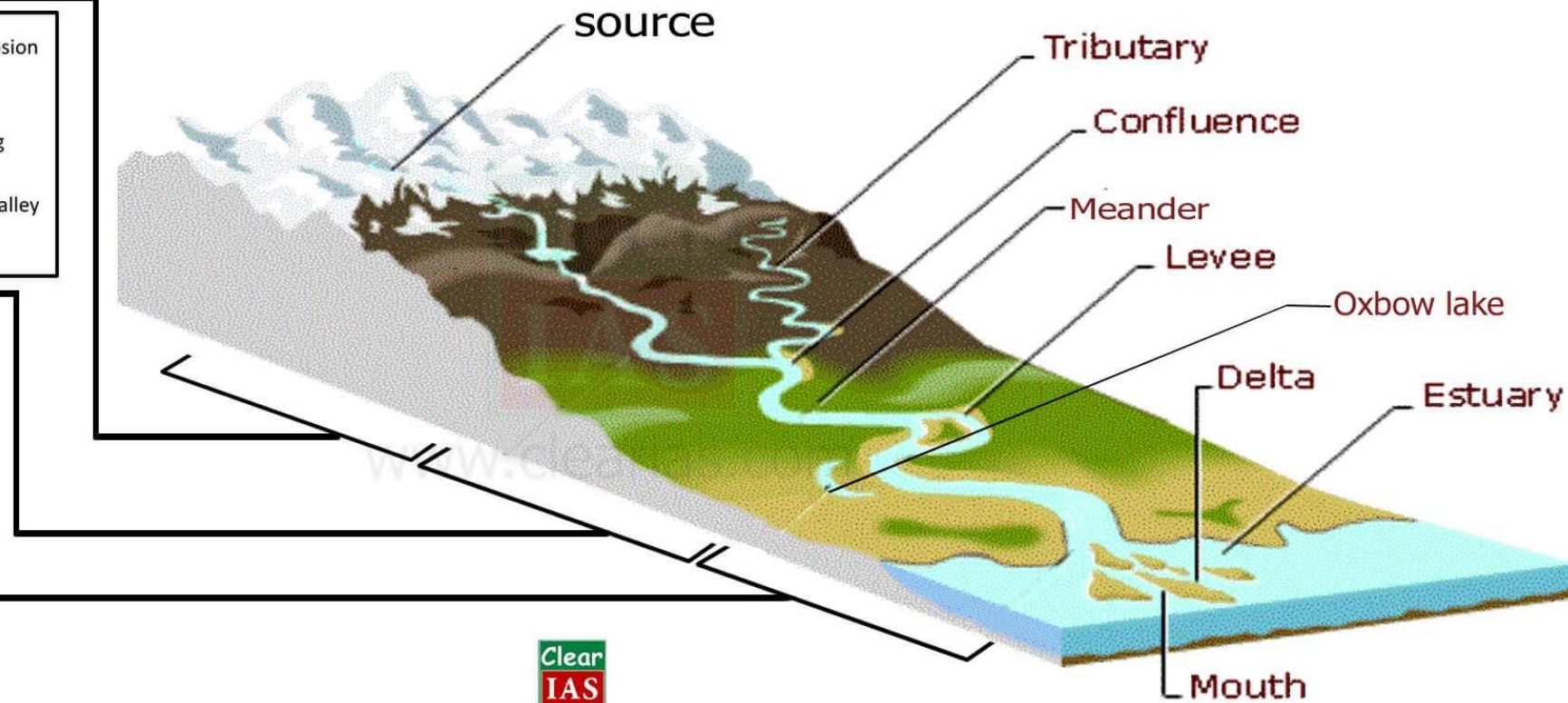


Middle Course

flows with moderate velocity
vertical and lateral erosion
plenty of streams
meanders, alluvial fans etc.

Lower Course

flows with low velocity
lateral erosion dominates &
wide channels
extensive flood plains
braided channels,
levees, oxbow lakes, meanders, deltas etc.

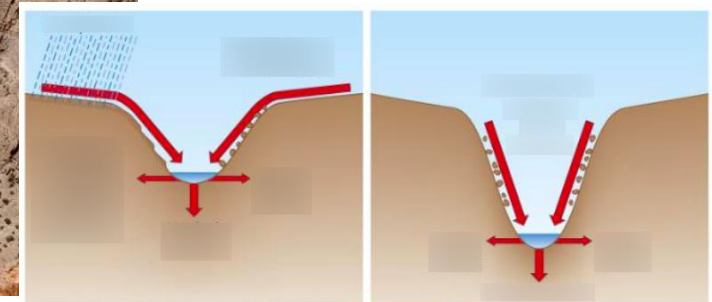
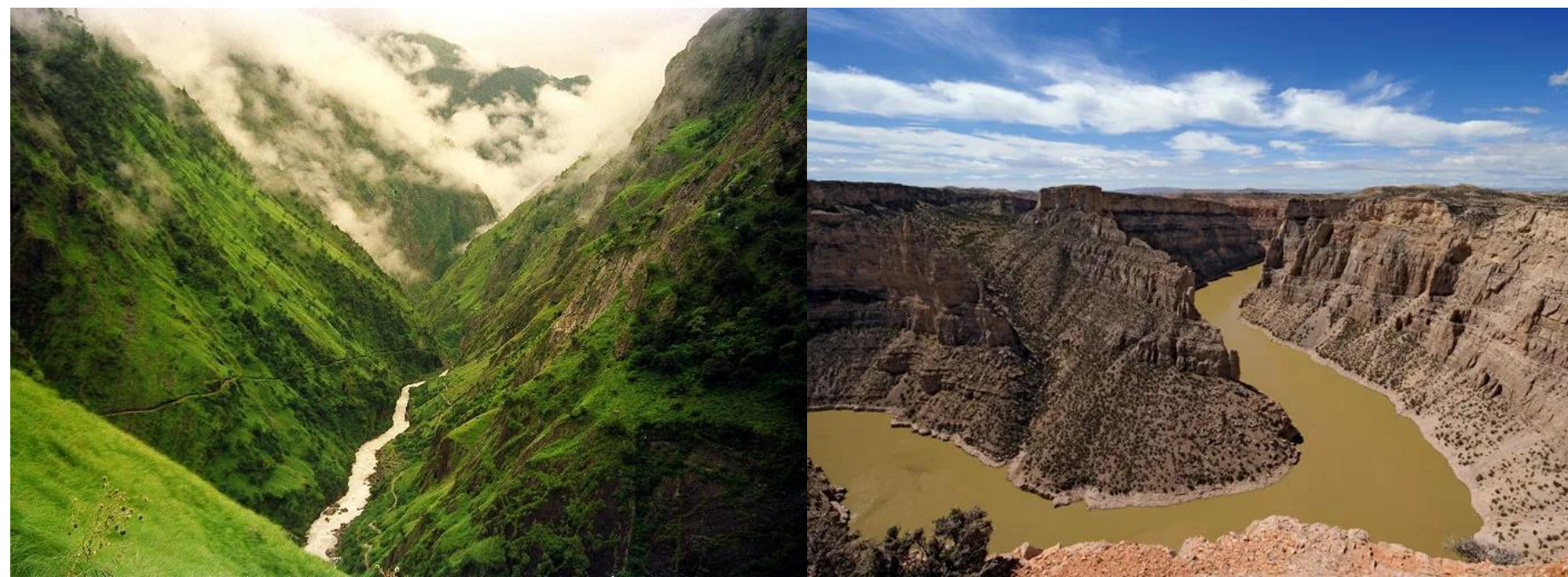


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Stages of a River

Geomorphic structure formed due to erosion of River



- Valleys start as small and narrow rills. These rills will progressively develop into long and wide gullies.
- The gullies will again deepen, widen and lengthen to give rise to valleys.
- The valley types depend upon the type and structure of rocks in which they form.
- Depending upon sizes and shapes, several types of valleys like V-shaped valley, gorge, canyon, etc. can be recognized.
- A gorge is a deep valley with very steep to straight sides. It is almost equal in width at its top as well as its bottom.
- A canyon is characterized by steep step-like side slopes and might be as deep as a gorge. It is a variant of the gorge.
- A canyon is wider at its top than at its bottom. It is commonly formed in horizontal bedded sedimentary rocks and gorges form in hard rocks.

Potholes



Geomorphic structure formed due to erosion of River

Potholes are cylindrical holes drilled into the bed of a river that varies in depth and diameter from a few centimetres to several metres. They are found in the upper course of a river where it has enough potential energy to erode vertically and its flow is turbulent.

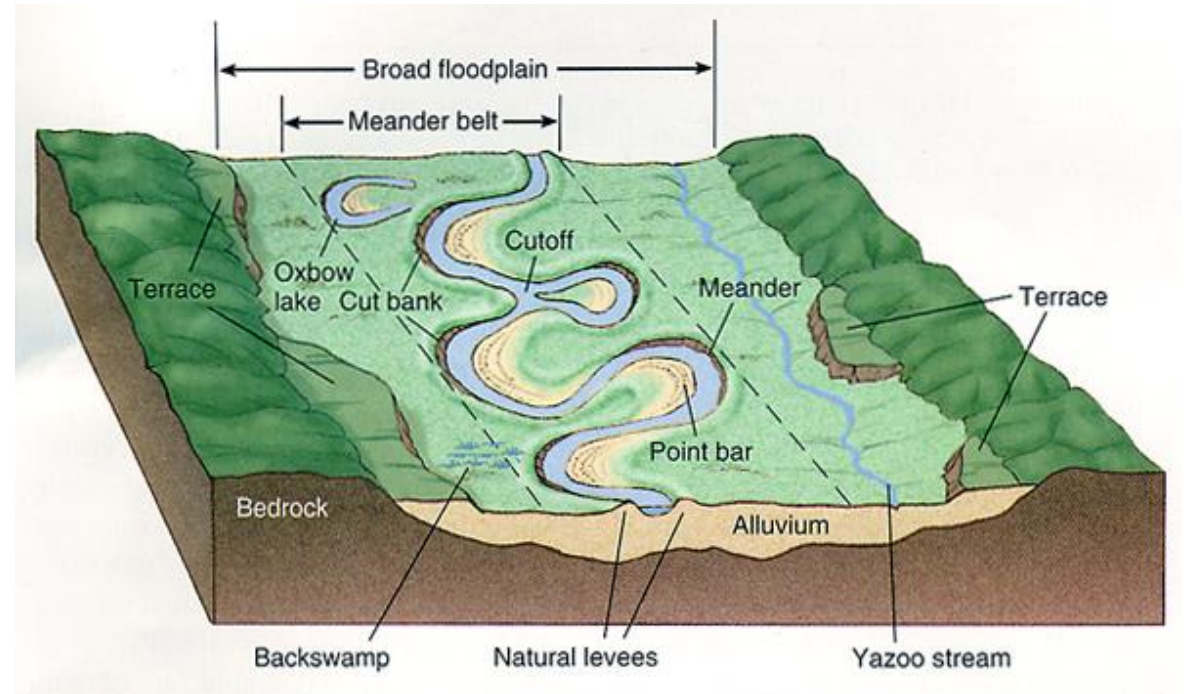
Cascade



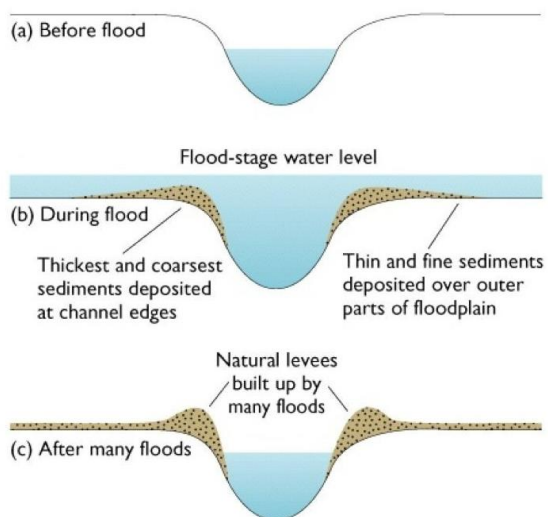
A **waterfall** is a point in a river or stream where water flows over a vertical drop or a series of steep drops.



Geomorphic structure formed by River



- **Meanders** are loop-like channel patterns develop over the flood and delta plains.
- The concave bank of a meander is known as cut-off bank and the convex bank is known as a slip-off
- As meanders grow into deep loops, the same may get cut-off due to erosion at the inflection point and are left as oxbow lakes.
- For large rivers, the sediments deposited in a linear fashion at the depositional side of a meander are called as **Point Bars or Meander Bars**.
- A riverbed made of river deposits is the active flood plain and the flood plain above the bank of the river is the inactive flood plain.
- **Natural levees** are found along the banks of large rivers. They are low, linear and parallel ridges of coarse deposits along the banks of a river.
- The levee deposits are coarser than the deposits spread by flood water away from the river.





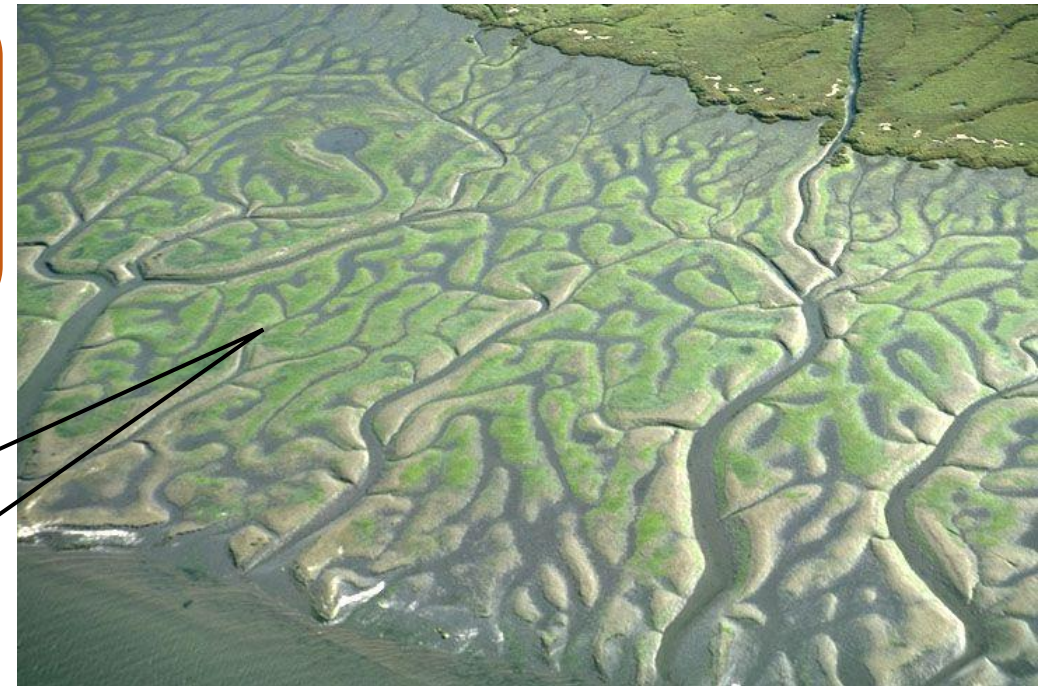
An **alluvial fan** is an accumulation of sediments that fans outwards from a concentrated source of sediments, such as a narrow canyon emerging from an escarpment.

A **braided river**, or braided channel, consists of a network of river channels separated by small, often temporary, islands called braid bars



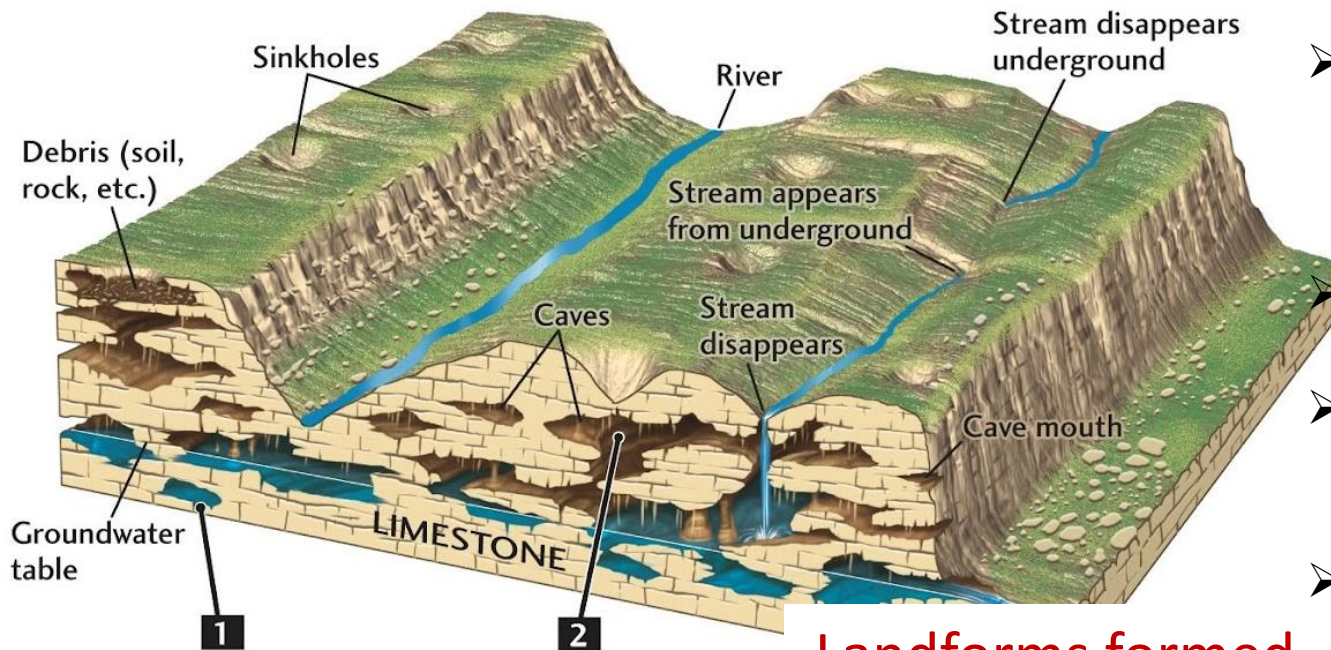
A **river delta** is a landform shaped like a triangle, created by deposition of sediment that is carried by a river and enters slower-moving or stagnant water

Tidal flats are coastal wetlands that form in intertidal areas where sediments have been deposited by tides or rivers.



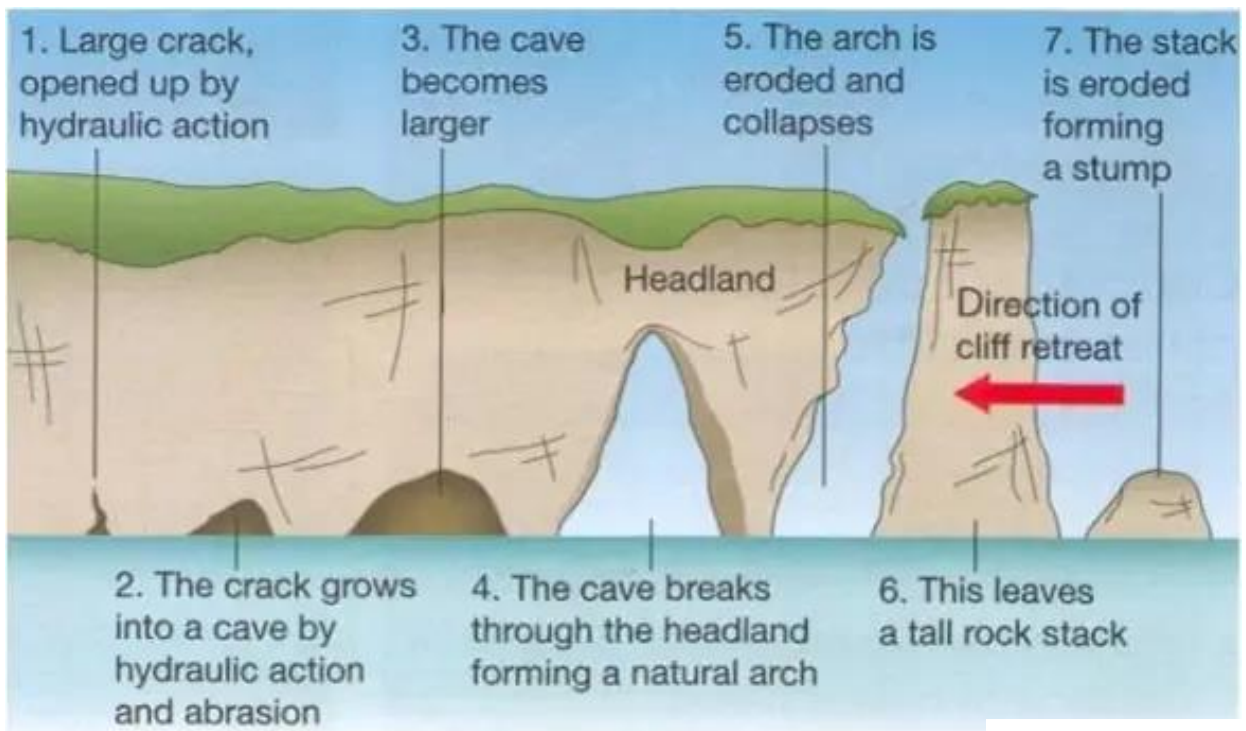


Minerve, Hérault, France.

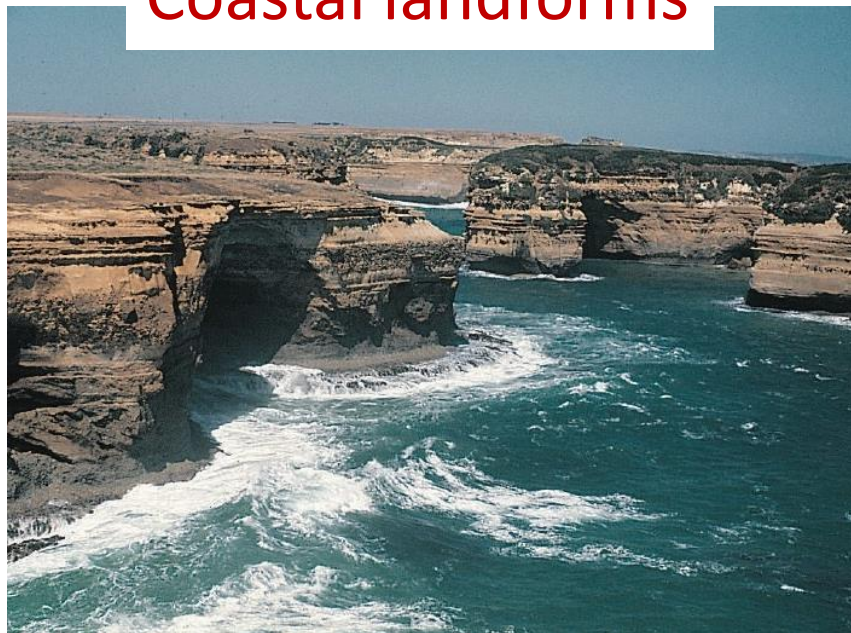


Landforms formed by Groundwater

- Karst is a topography formed from the dissolution of soluble rocks such as limestone, dolomite, and gypsum. It is characterized by underground drainage systems with sinkholes and caves.
- It has also been documented for more weathering-resistant rocks, such as quartzite, given the right conditions.
- A stalactite is a mineral formation that hangs from the ceiling of caves, hot springs, or man-made structures such as bridges and mines.
- A stalagmite is a type of rock formation that rises from the floor of a cave due to the accumulation of material deposited on the floor from ceiling drippings. Stalagmites are typically composed of calcium carbonate.



Coastal landforms



Geomorphic structure formed due to wind erosion



A **yardang** is a streamlined protuberance carved from bedrock or any consolidated or semiconsolidated material by the dual action of wind abrasion by dust and sand and deflation (the removal of loose material by wind turbulence.)



A **ventifact** (also wind-faceted stone, windkanter) is a rock that has been abraded, pitted, etched, grooved, or polished by wind-driven sand or ice crystals. These geomorphic features are most typically found in arid environments where there is little vegetation to interfere with aeolian particle transport, where there are frequently strong winds, and where there is a steady but not overwhelming supply of sand.

A **mushroom rock**, also called rock pedestal, or a pedestal rock, is a naturally occurring rock whose shape, as its name implies, resembles a mushroom.

Wind Depositional Structures

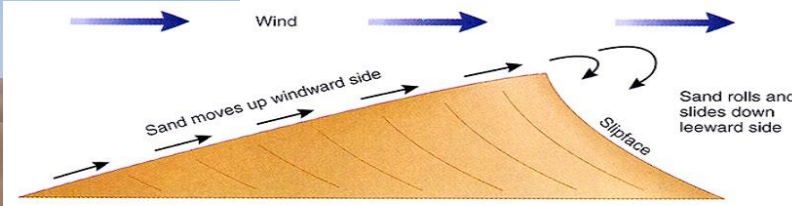


FIGURE 12.7
Cross section of a dune, showing sand moving up the gentle windward slope and falling off the steeper leeward slope.

Sand dunes in dessert

Longitudinal sand dunes

Barchan sand dunes

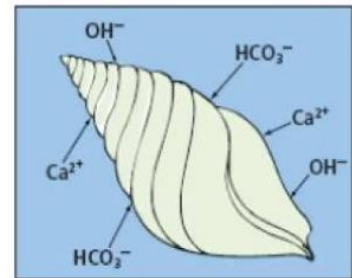
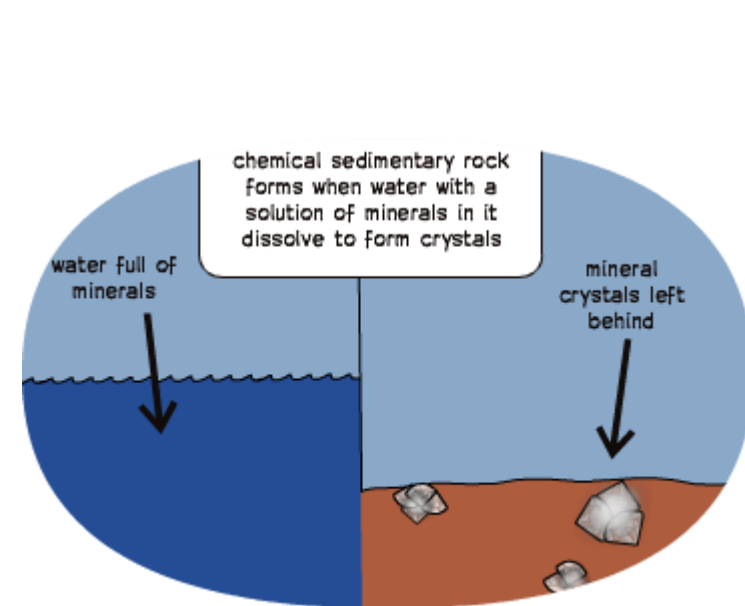


The West Temple mountain (2,380 m) in Zion National Park

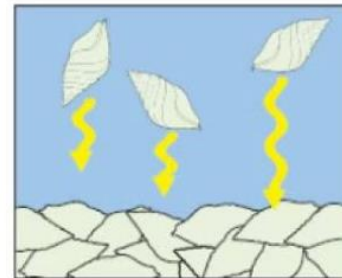
Sand dune, any accumulation of sand grains shaped into a mound or ridge by the wind under the influence of gravity.

Chemical sedimentary rocks

- Chemical sedimentary rocks and biochemical sedimentary rocks are derived from material (ions) that is carried in solution to lakes and seas.
- This material does not remain dissolved in water indefinitely.
- Under certain conditions, it precipitates (settles out) to form chemical sediments as a result of physical processes.
- An example of chemical sediments resulting from physical processes is the salt left behind as a body of saltwater evaporates.
- Precipitation may also occur indirectly through life processes of water-dwelling organisms that form materials called biochemical sediments.
- Many water-dwelling animals and plants extract dissolved mineral matter to form shells and other hard parts. After the organisms die, their skeletons may accumulate on the floor of a lake or an ocean.



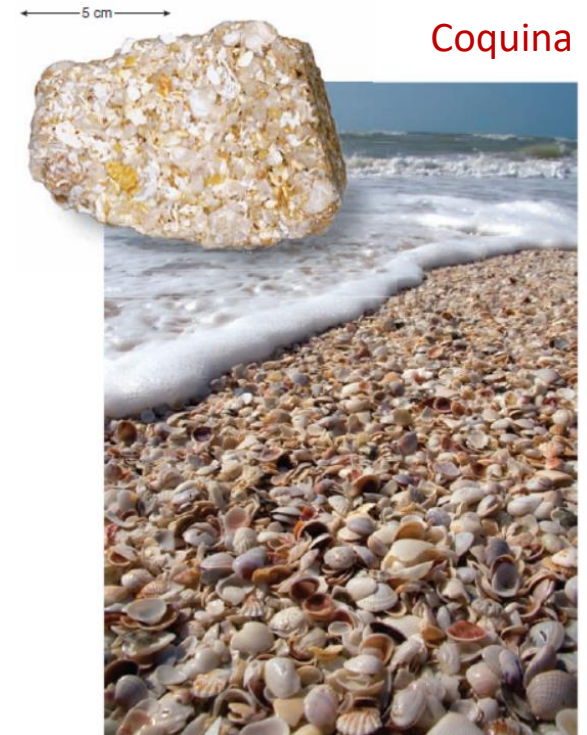
Organisms that live in lakes or oceans take chemicals from the water and produce the mineral calcium carbonate, CaCO_3 . They use the CaCO_3 to build their shells or skeletons.



When the organisms die, the hard remains that are made of CaCO_3 settle to the lake or ocean floor.



The shells of the dead organisms pile up. Eventually, the layers are compacted and cemented to form limestone.



Chemical, Biochemical, and Organic Sedimentary Rocks

Composition	Texture	Rock Name	
Calcite, CaCO ₃	Nonclastic: Fine to coarse crystalline	Crystalline Limestone	
	Nonclastic: Microcrystalline calcite	Microcrystalline Limestone	
	Nonclastic: Fine to coarse crystalline	Travertine	
	Clastic: Visible shells and shell fragments loosely cemented	Coquina	
	Clastic: Various size shells and shell fragments cemented with calcite cement	Fossiliferous Limestone	
Biochemical Limestone	Clastic: Microscopic shells and clay	Chalk	

Limestone, an abundant sedimentary rock, is composed chiefly of the mineral calcite (CaCO₃). Nearly 90 percent of limestone is formed from biochemical sediments secreted by marine organisms, and the remaining amount consists of chemical sediments that precipitated directly from seawater.

Inorganic limestone forms when chemical changes or high water temperatures increase the concentration of calcium carbonate to the point that it precipitates. Travertine, the type of limestone that decorates caverns. Groundwater is the source of travertine that is deposited in caves. As water drops reach the air in a cavern, some of the carbon dioxide dissolved in the water escapes, causing calcium carbonate to precipitate.

One easily identified biochemical limestone is coquina, a coarse rock composed of loosely cemented shells and shell fragments.

Another less obvious but familiar example is chalk, a soft, porous rock made up almost entirely of the hard parts of microscopic organisms that are no larger than the head of a pin.



Chemical, Biochemical, and Organic Sedimentary Rocks

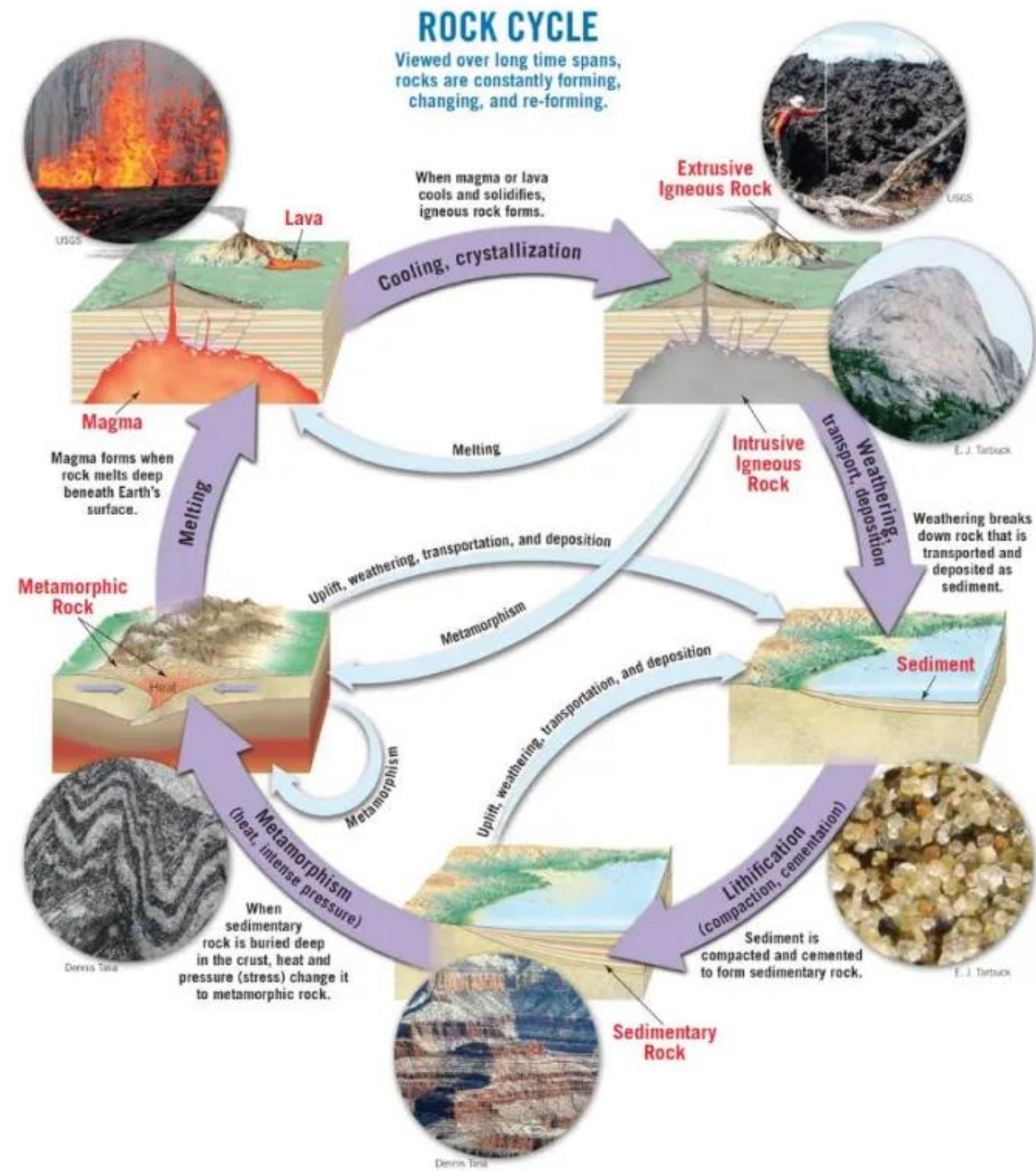
Composition	Texture	Rock Name
Quartz, SiO_2	Nonclastic: Very fine crystalline	Chert (light colored) 
Gypsum $\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$	Nonclastic: Fine to coarse crystalline	Rock Gypsum 
Halite, NaCl	Nonclastic: Fine to coarse crystalline	Rock Salt 
Altered plant fragments (organic)	Nonclastic: Fine-grained organic matter	Bituminous Coal 

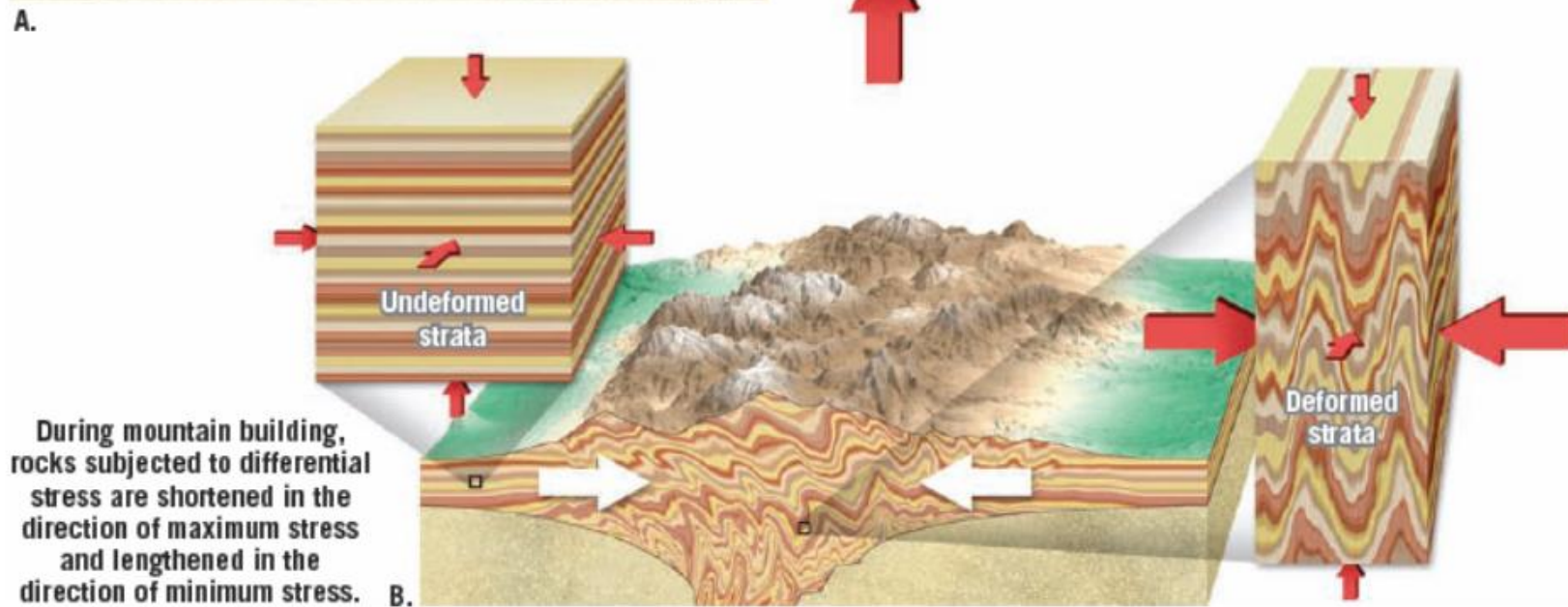
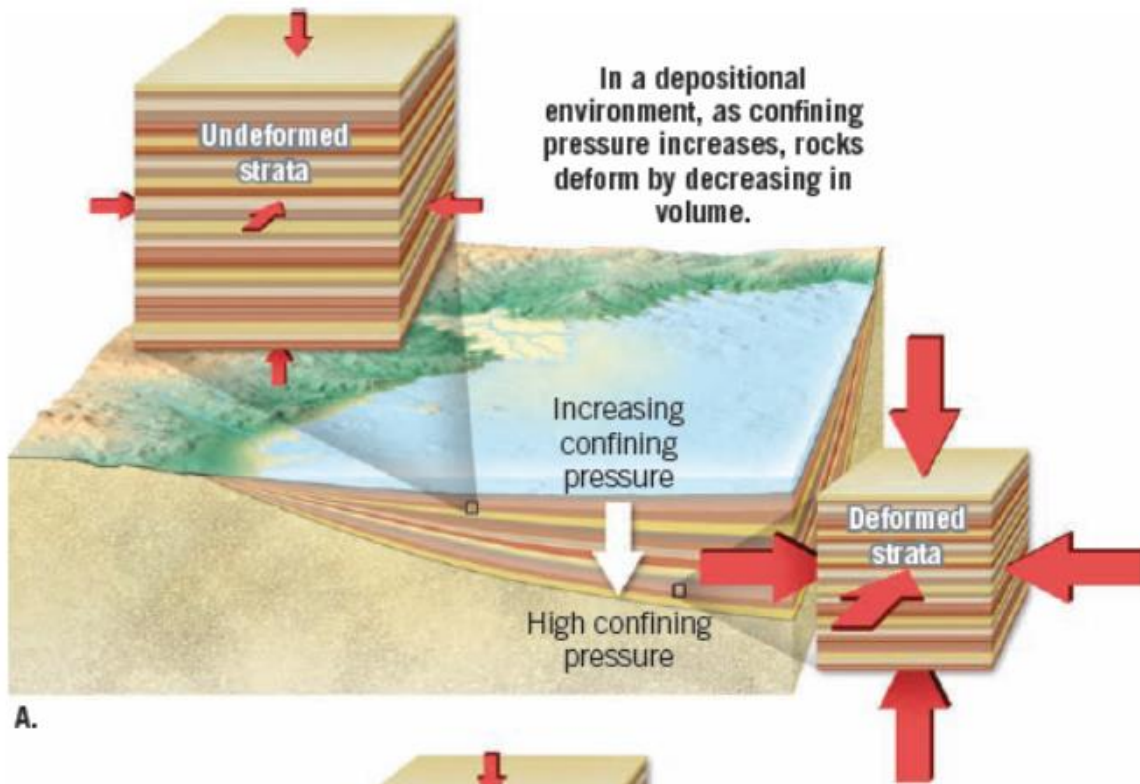
Dissolved silica (SiO_2) precipitates to form varieties of microcrystalline quartz. Sedimentary rocks composed of microcrystalline quartz include chert (light color), flint (dark), jasper (red), and agate (banded). These chemical sedimentary rocks may have either an inorganic or biochemical origin, but the mode of origin is usually difficult to determine.

Evaporation causes minerals to precipitate from water. Such minerals include halite, the chief component of **rock salt**, and **gypsum**, the main ingredient of rock gypsum. Both materials have significant commercial importance. Halite is familiar to everyone as the common salt used in cooking and seasoning foods. Gypsum is the basic ingredient of plaster of Paris. This material is used most extensively in the construction industry for “drywall” and plaster.

Coal, in contrast to sedimentary rocks that are rich in calcite or silica, consists mostly of organic matter.

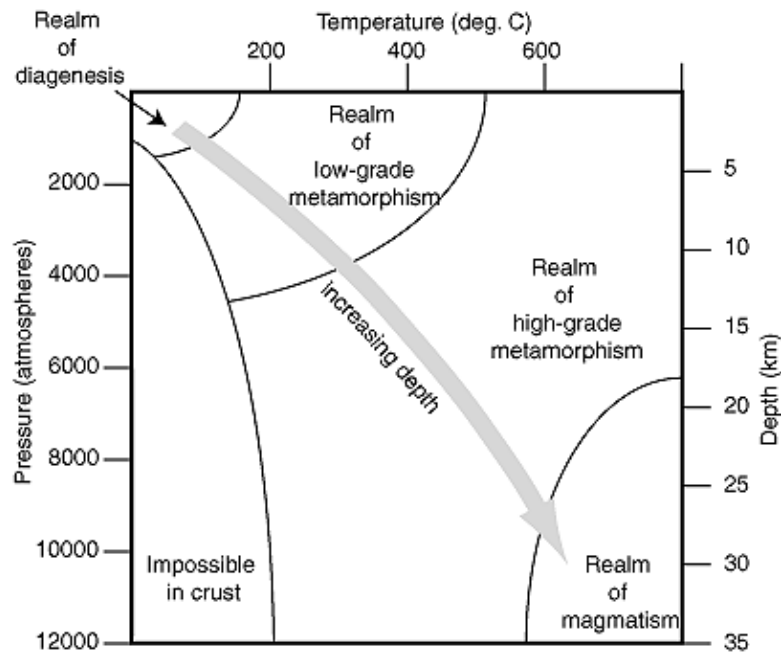
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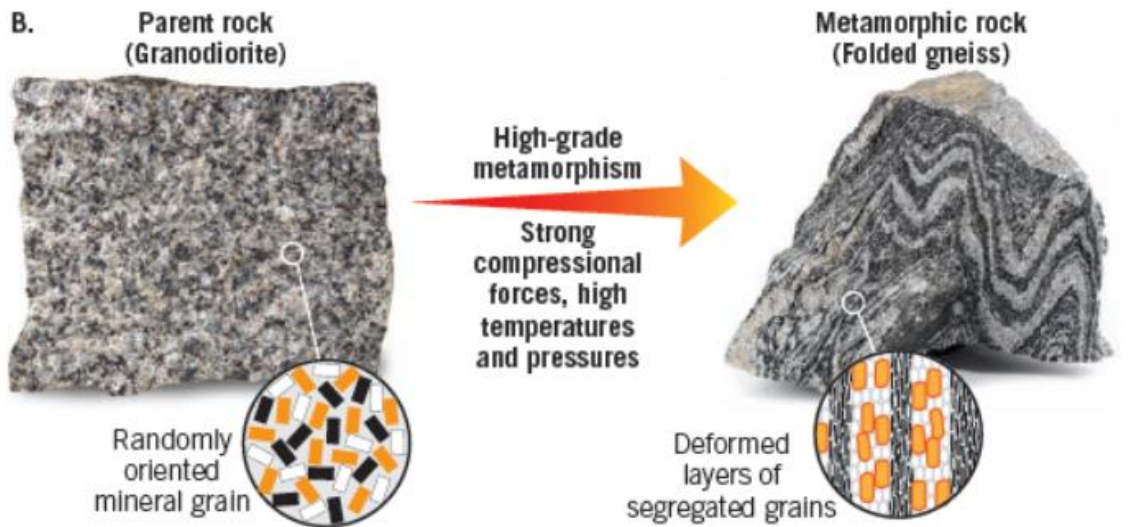
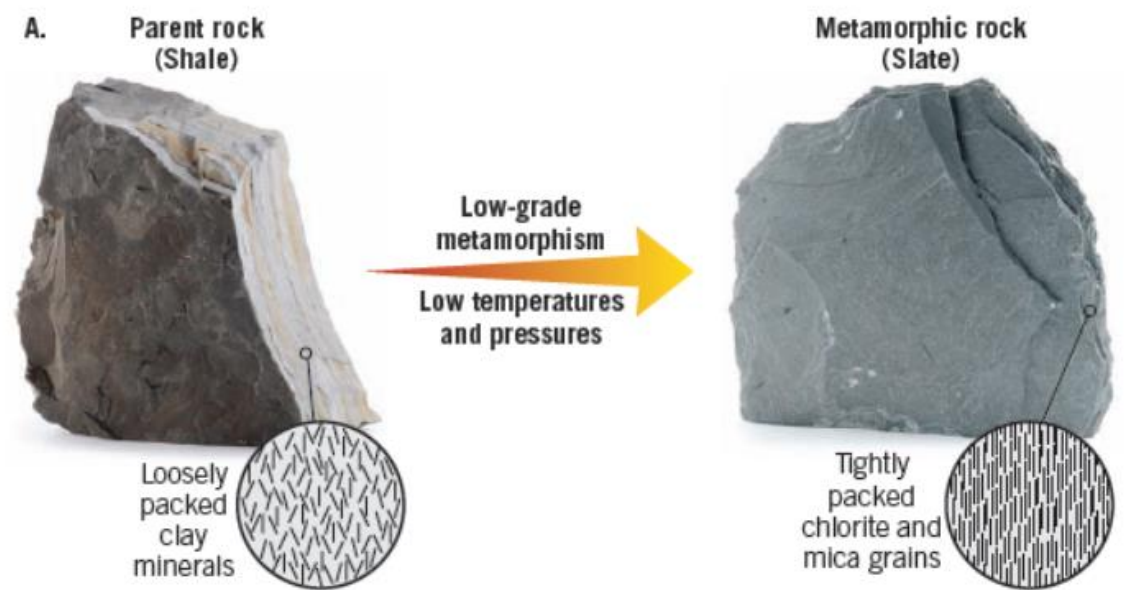


Metamorphism, which means “to change form,” is a process that leads to changes in the mineralogy, texture (for example, grain size), and sometimes chemical composition of rocks.

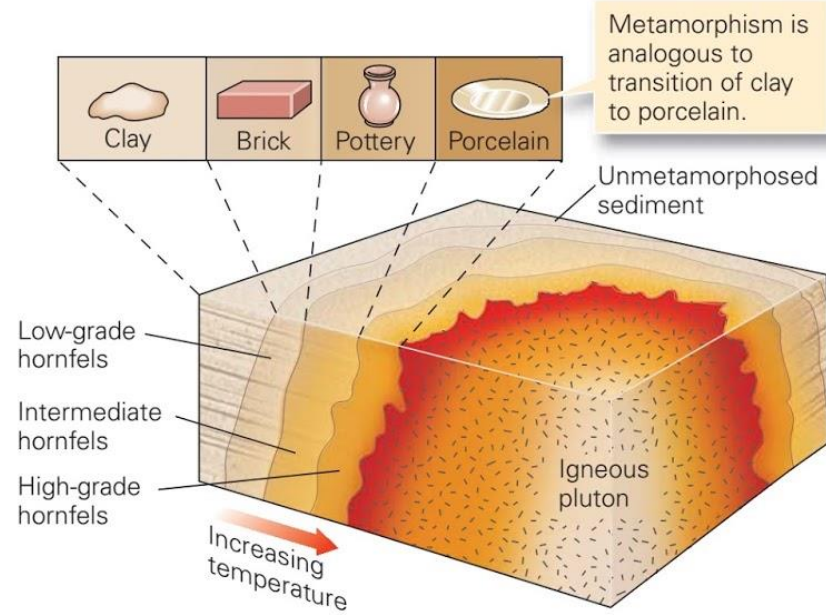
- Metamorphism takes place when preexisting rock is subjected to a **physical or chemical environment** that is significantly different from that in which it initially formed.
- In response to these new conditions, the rock gradually changes until a state of equilibrium with the new environment is reached.
- Most metamorphic changes occur at the elevated temperatures and pressures that exist in the zone beginning a few kilometers below Earth's surface and extending into the mantle.
- Metamorphism often progresses incrementally, from slight changes (low-grade metamorphism) to substantial changes (high-grade metamorphism).



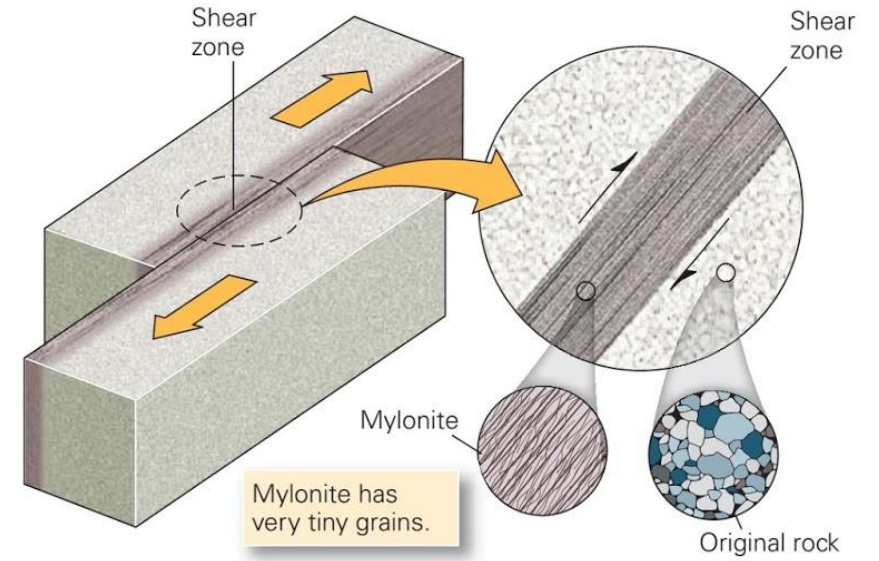
A. Low-grade metamorphism illustrated by the transformation of the common sedimentary rock shale to the more compact metamorphic rock slate.



B. High-grade metamorphic environments obliterate the existing texture and often change the mineralogy of the parent rock. High-grade metamorphism occurs at temperatures that approach those at which rocks melt. (Photos by Dennis Tasa)

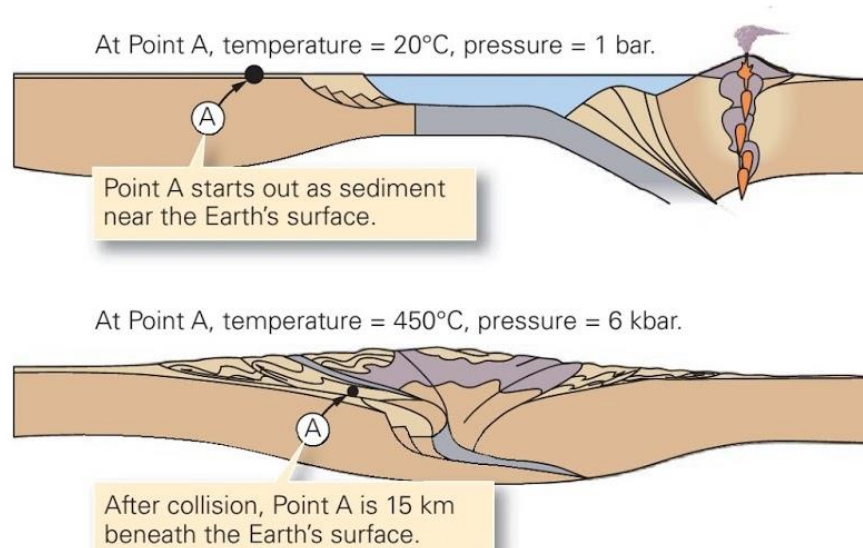


(a) Heat radiated from a large pluton can produce a metamorphic aureole, in which hornfels develops. Grade decreases progressively away from the pluton contact.

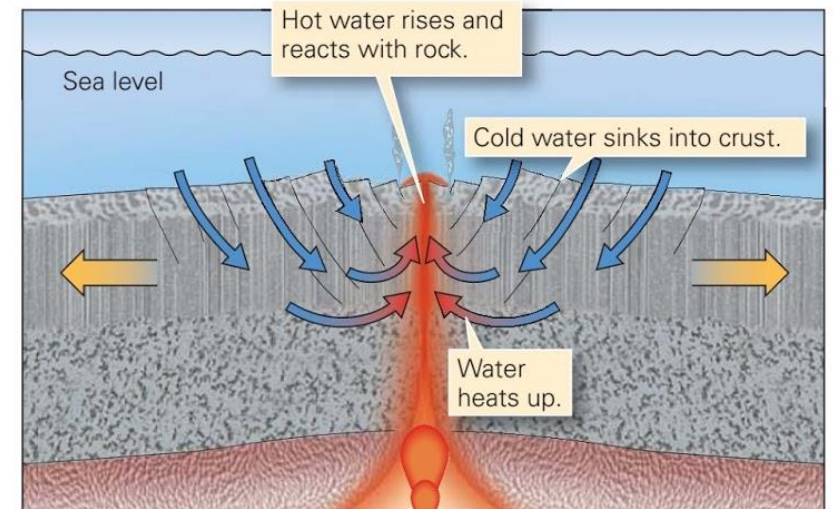


(b) Shearing of a rock under plastic conditions causes original crystals to divide into tiny crystals without breaking to form a mylonite. Mylonite has strong foliation.

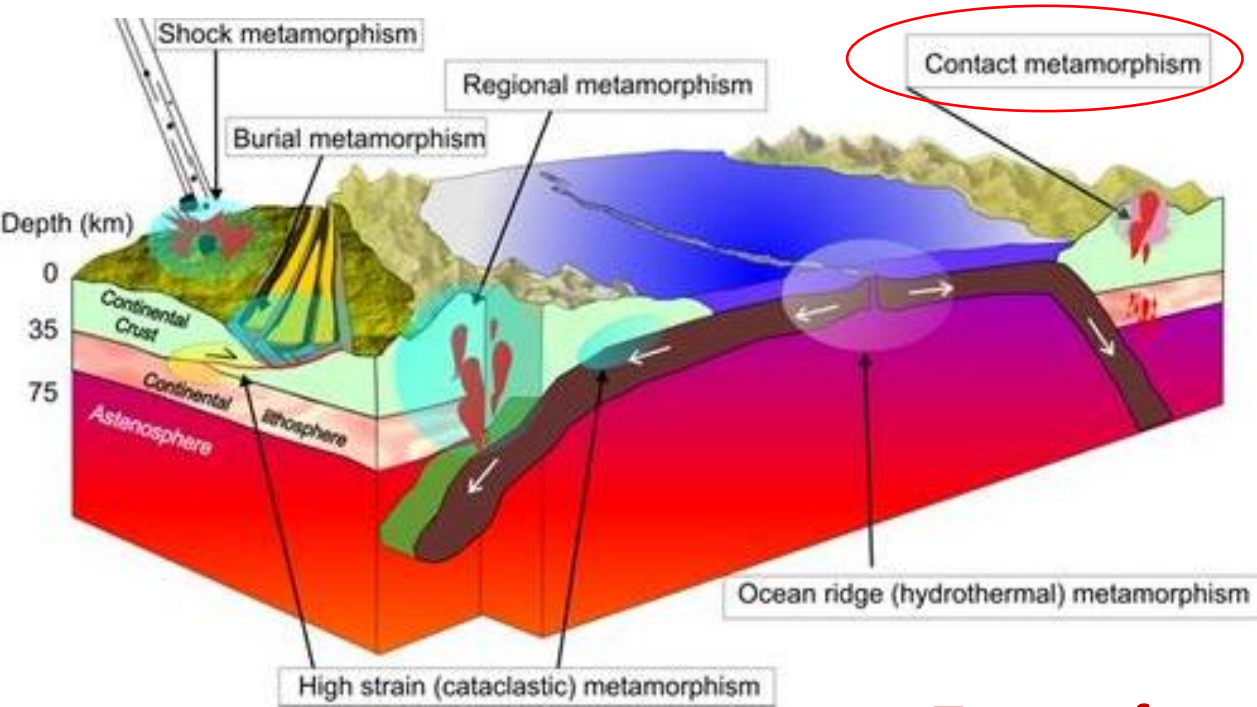
Where we get to see metamorphism?



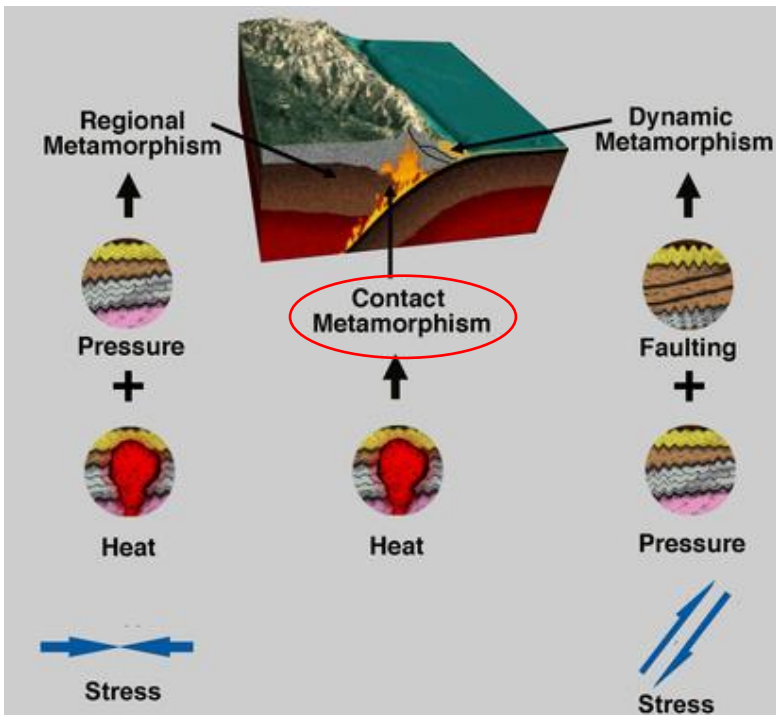
(c) Dynamothermal metamorphism happens when one part of the crust shoves over another part, so that rocks once near the surface end up at great depth.



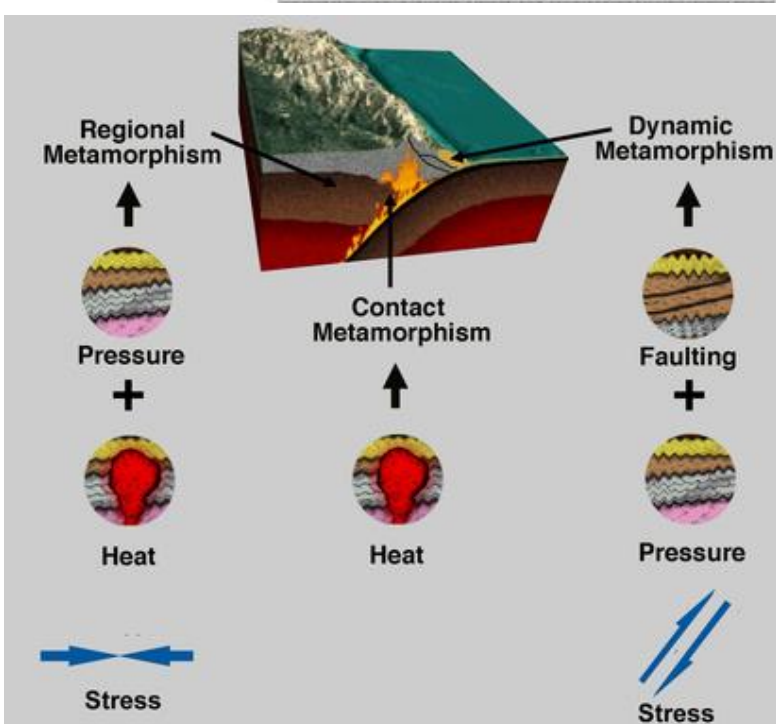
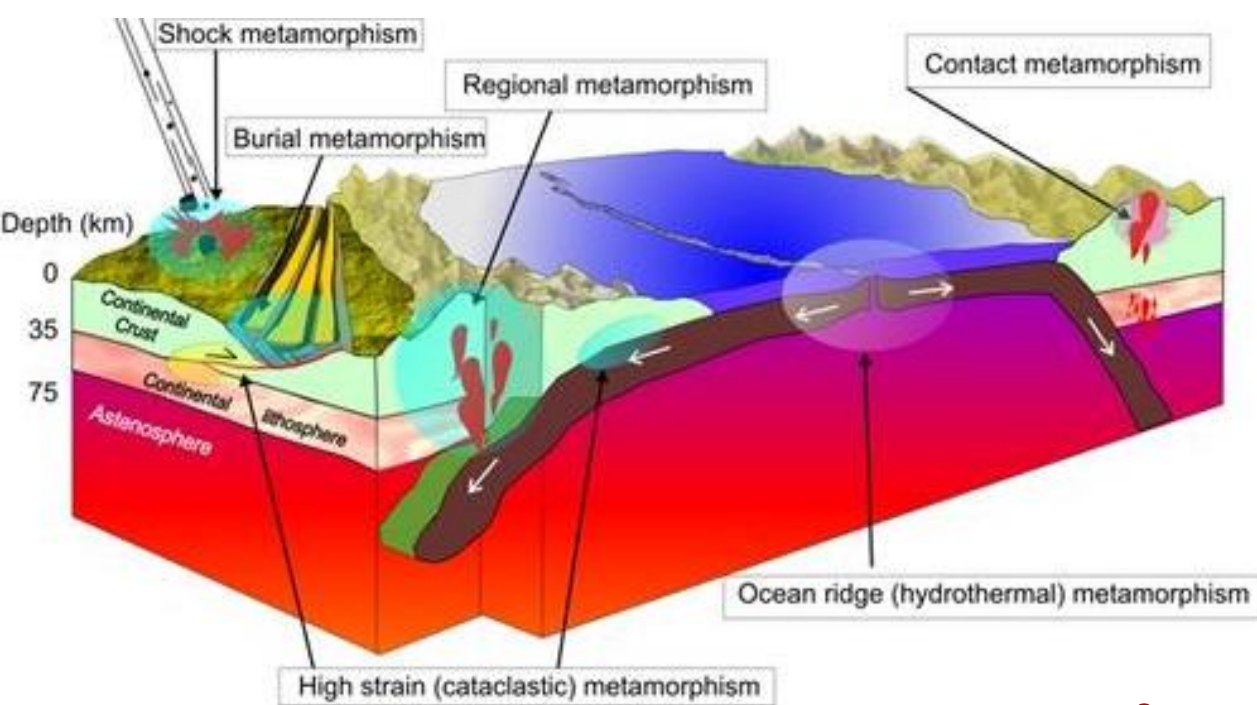
(d) The heat of rising magma at a ridge axis heats water, which then convects. The hot water reacts with the crust and forms metamorphic minerals.



Types of Metamorphism



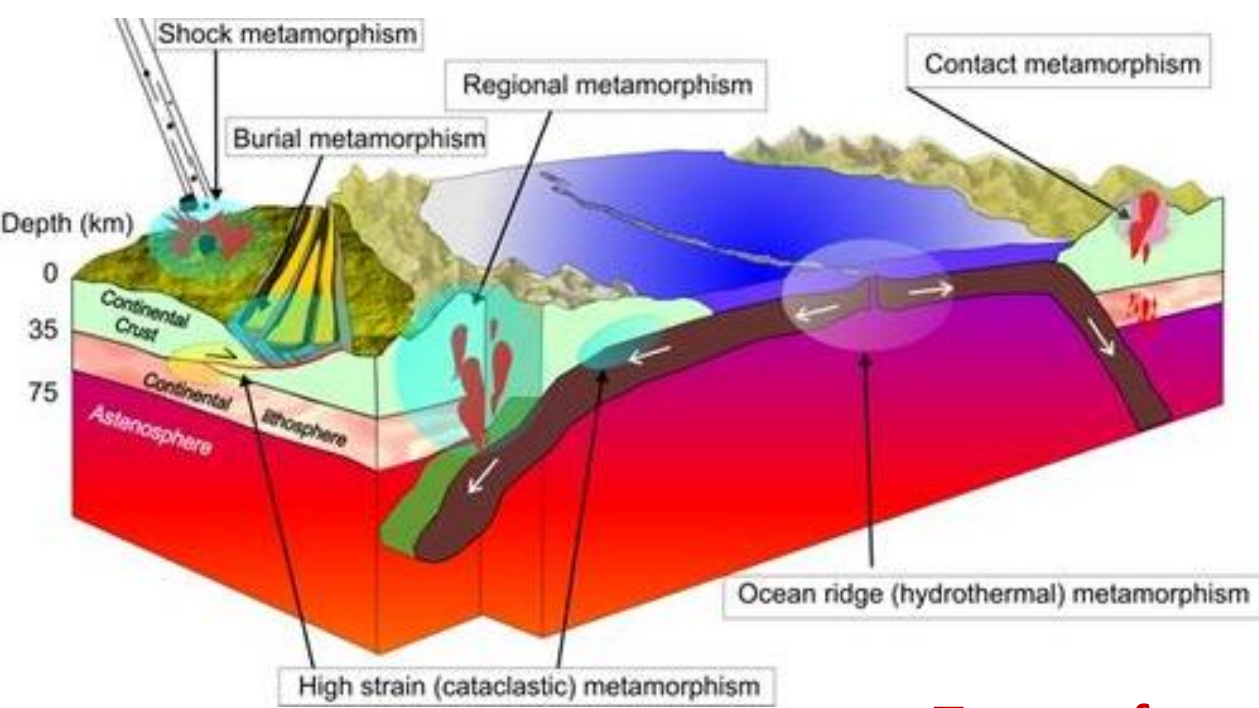
- **Thermal energy** (heat) is the most important factor driving metamorphism.
- It triggers chemical reactions that result in the recrystallization of existing minerals and the formation of new minerals.
- Thermal energy for metamorphism comes mainly from two sources.
- Rocks experience a rise in temperature when they are intruded by magma rising from below. This is called **contact, or thermal, metamorphism**. In this situation, the adjacent host rock is “baked” by the emplaced magma.
- By contrast, rocks that formed at Earth’s surface will experience a **gradual increase in temperature and pressure** as they are taken to greater depths.
- In the upper crust, this increase in temperature averages about 25°C per kilometer.
- When buried to a depth of about 8 kilometers, (T 150°- 200°C), clay minerals tend to become unstable and begin to recrystallize into other minerals, such as chlorite and muscovite, that are stable in this environment. (Chlorite is a mica-like mineral formed by the metamorphism of iron- and magnesium- rich silicates.)
- However, many silicate minerals, particularly those found in crystalline igneous rocks—quartz and feldspar, for example—remain stable at these temperatures. Thus, metamorphic changes in these minerals require much higher temperatures in order to recrystallize.



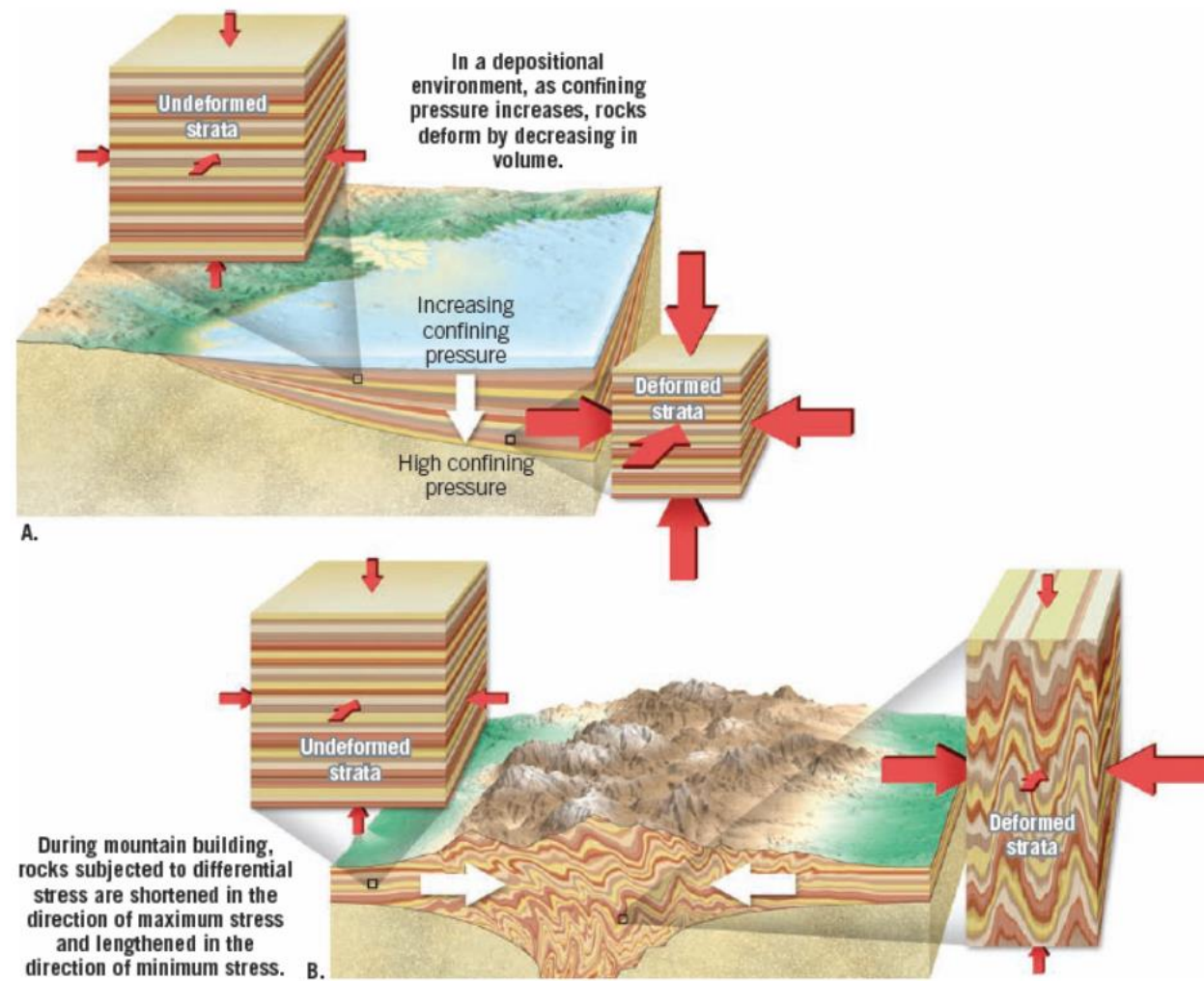
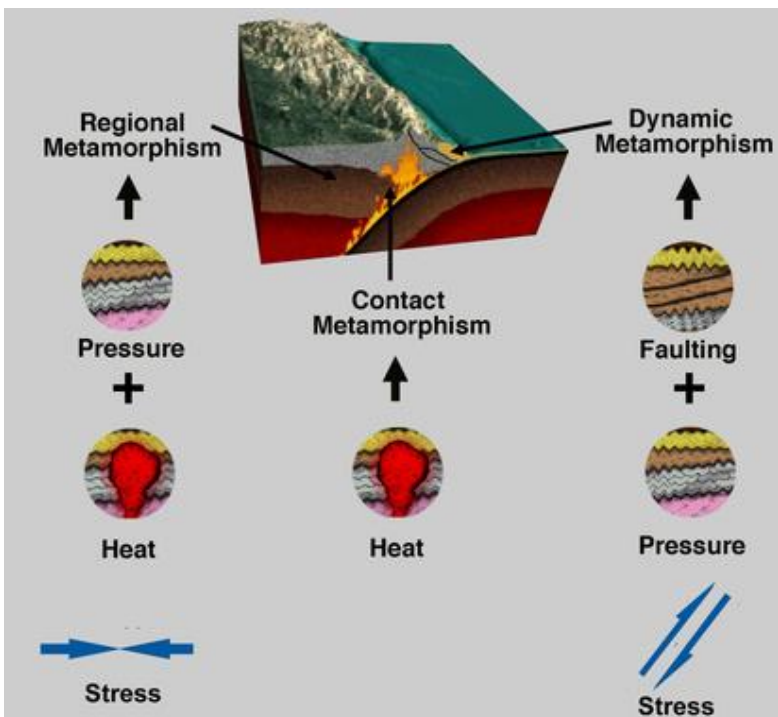
Types of Metamorphism

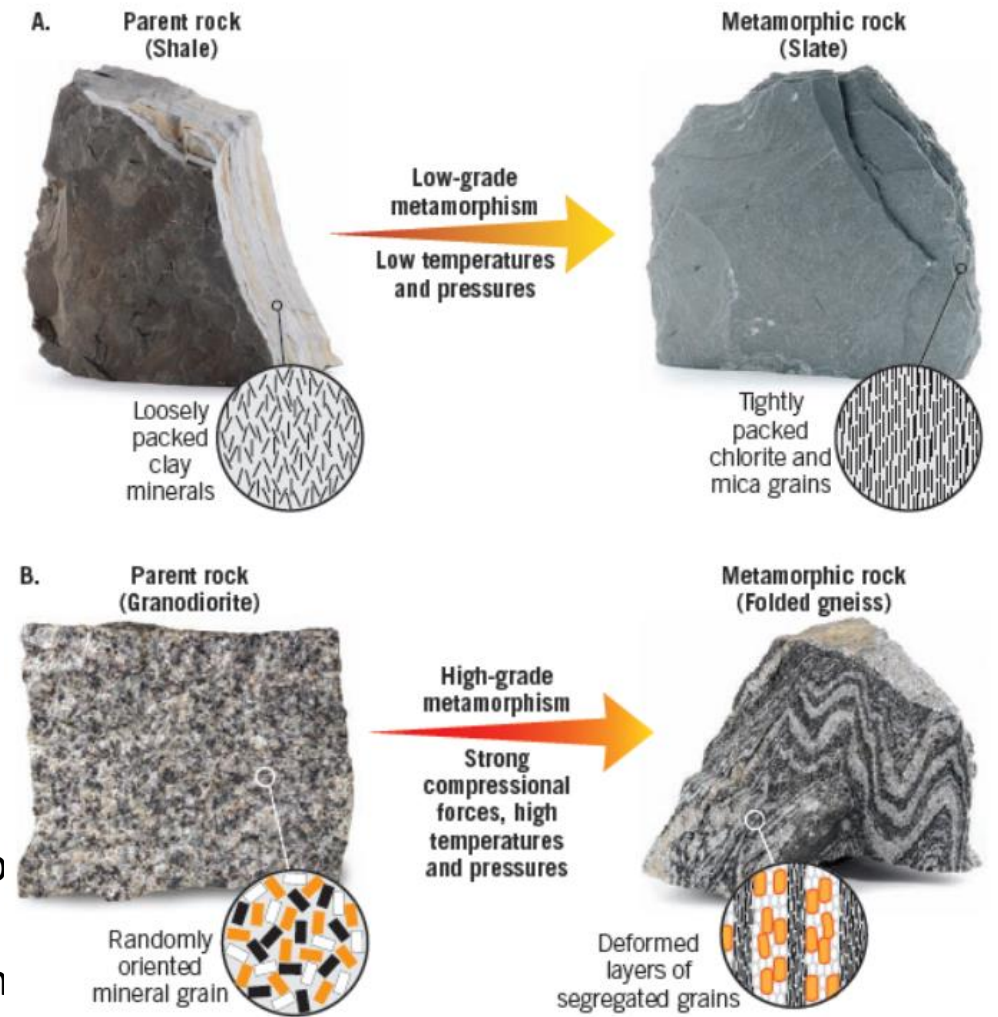
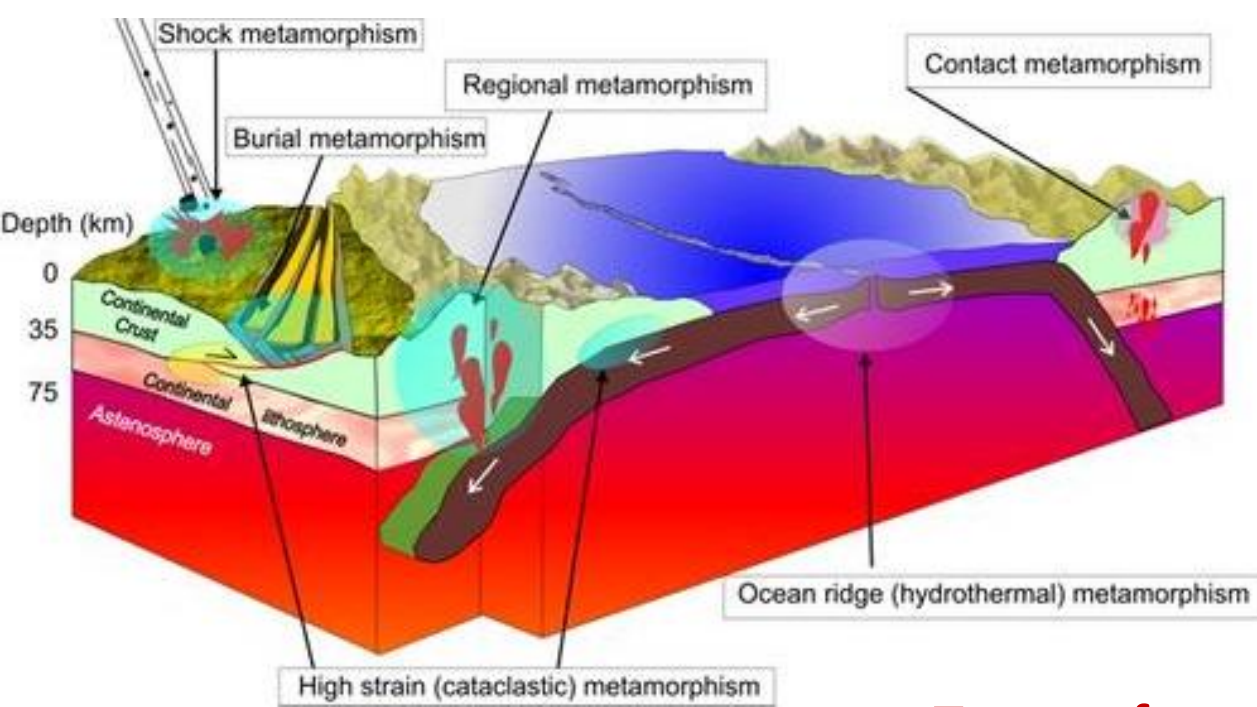
- During episodes of mountain building, large rock bodies become highly crumpled and metamorphosed.
- The forces that generate mountains are unequal in different directions and are called differential stress.
- Unlike **confining pressure**, which “squeezes” rock equally in all directions, differential stresses are greater in one direction than in others.
- The deformation caused by **differential stresses** plays a major role in developing metamorphic textures.
- When rocks exhibit ductile behavior, their mineral grains tend to flatten and elongate when subjected to differential stress

- **Confining Pressure and Differential Stress:** Metamorphic Agents Pressure, like temperature, increases with depth as the thickness of the overlying rock increases.
- Buried rocks are subjected to confining pressure— similar to water pressure in that the forces are equally applied in all directions.
- The deeper you go in the ocean, the greater the confining pressure. The same is true for buried rock.
- Confining pressure causes the spaces between mineral grains to close, producing a more compact rock that has greater density.
- Further, at great depths, confining pressure may cause minerals to recrystallize into new minerals that display more compact crystalline forms.



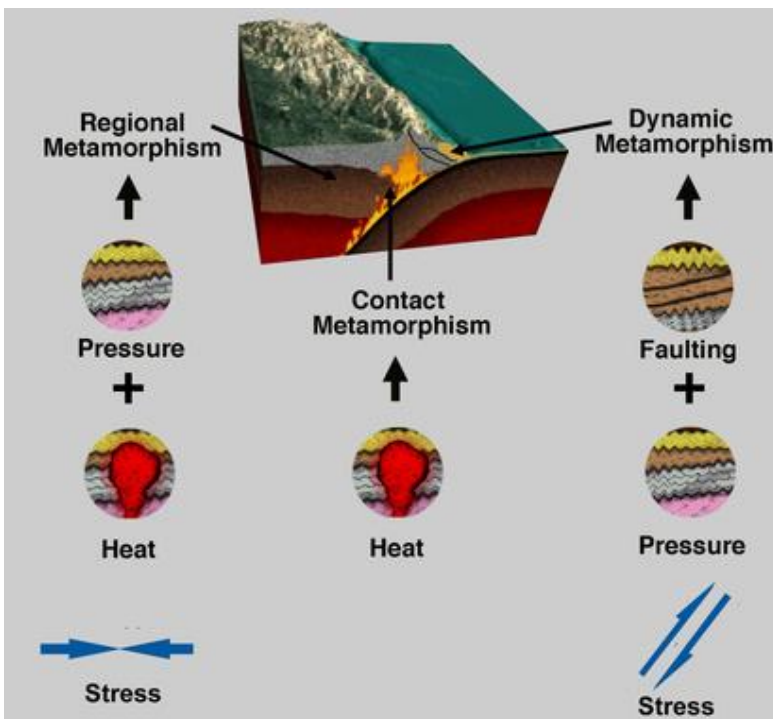
Types of Metamorphism

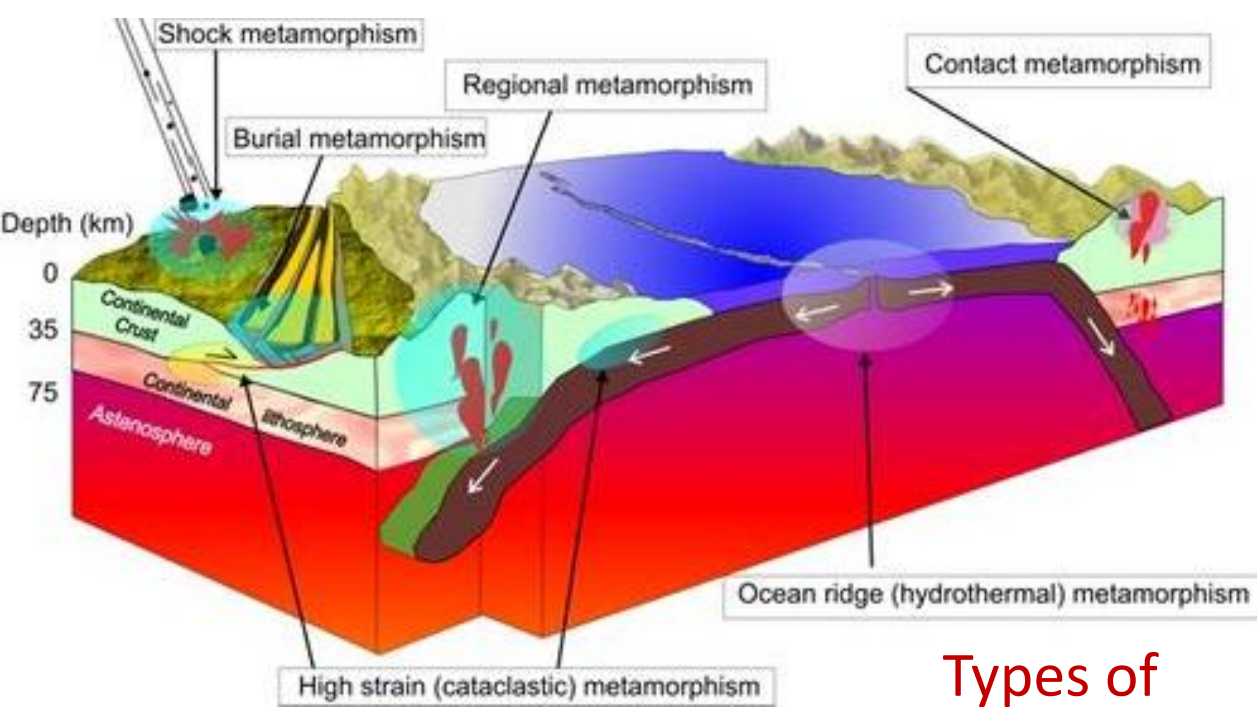




Types of Metamorphism

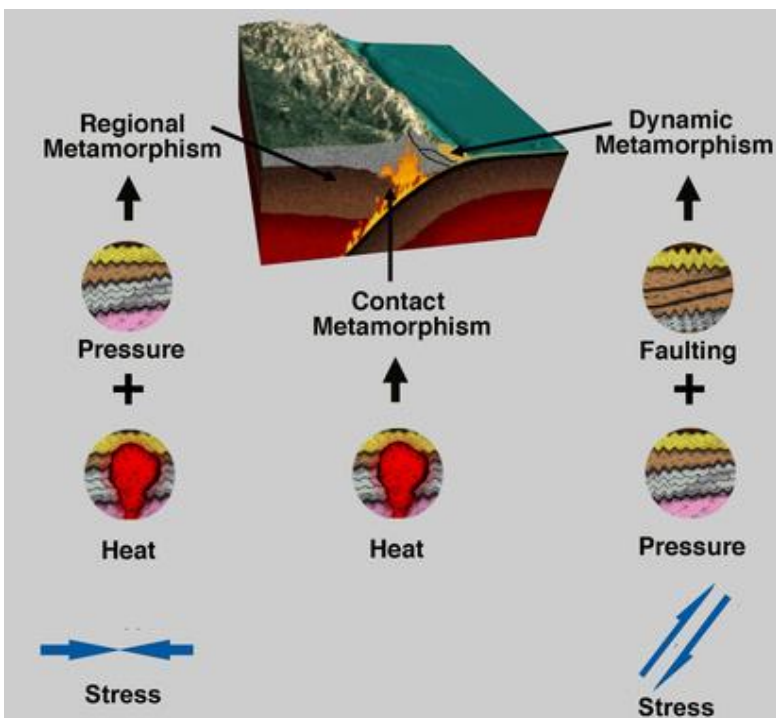
- During episodes of mountain building and metamorphism.
- The forces that generate mountains are called differential stress.
- Unlike confining pressure, which “squeezes” rock equally in all directions, differential stresses are greater in one direction than in others.
- The deformation caused by **differential stresses** plays a major role in developing metamorphic textures.
- When rocks exhibit ductile behavior, their mineral grains tend to flatten and elongate when subjected to differential stress.



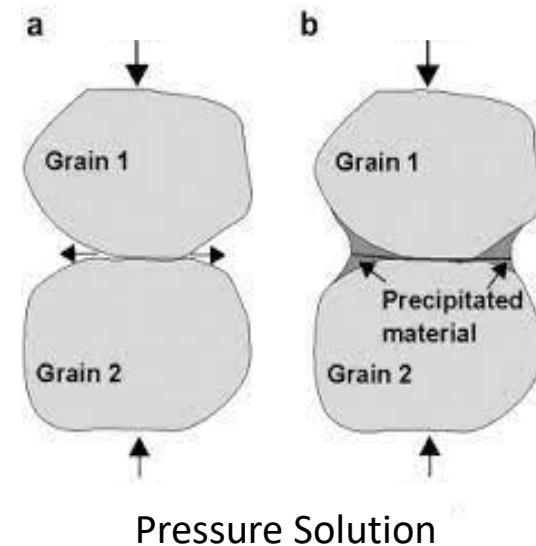


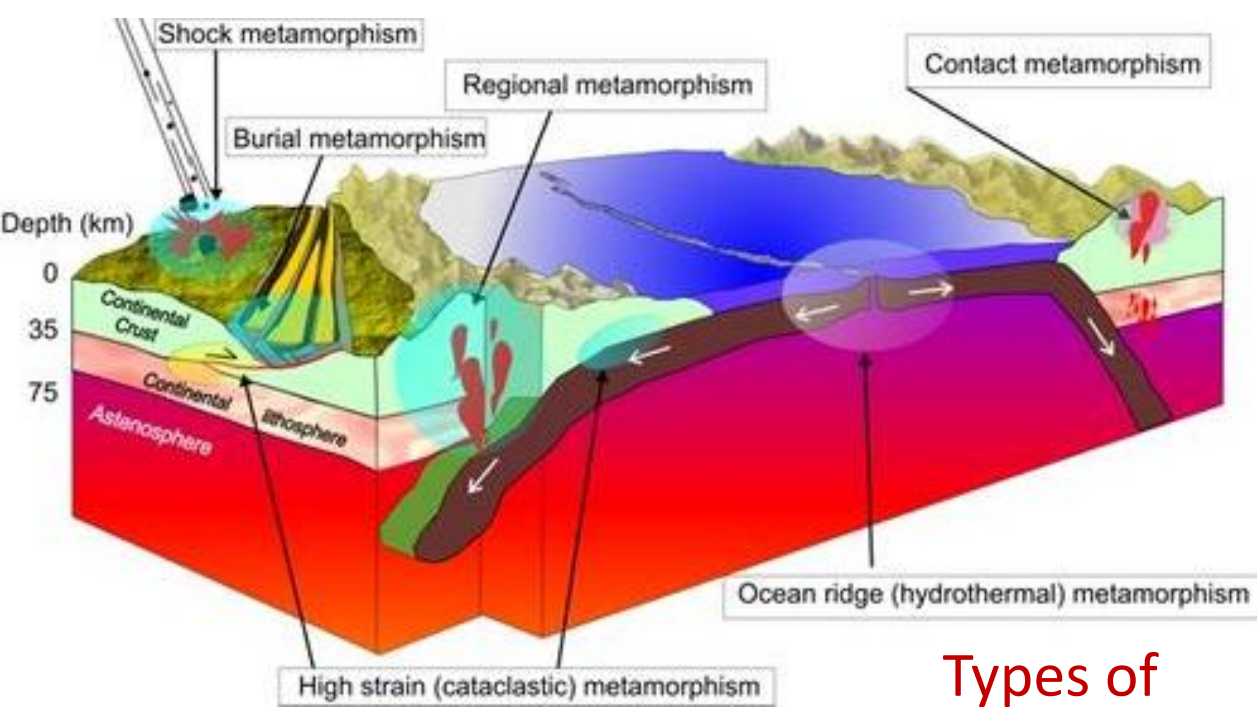
Types of Metamorphism

- **Chemically Active Fluids** Ion-rich fluids composed mainly of water and other volatiles (materials that readily change to gases at surface conditions) are believed to play an important role in some types of metamorphism.
- Fluids that surround mineral grains act **as catalysts to promote recrystallization by enhancing ion migration**. In progressively hotter environments, these ion-rich fluids become correspondingly more reactive.
- Chemically active fluids can produce **two types** of metamorphism, explained below.
- The first type changes the arrangement and **shape of mineral grains** within a rock.
- The second type **changes the rock's chemical composition**.



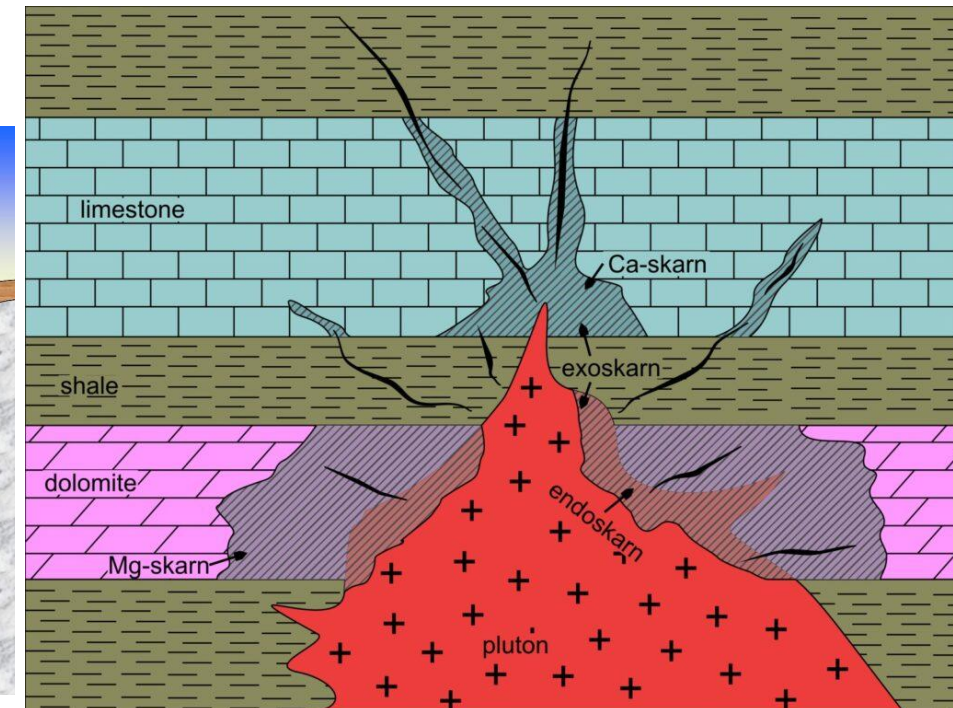
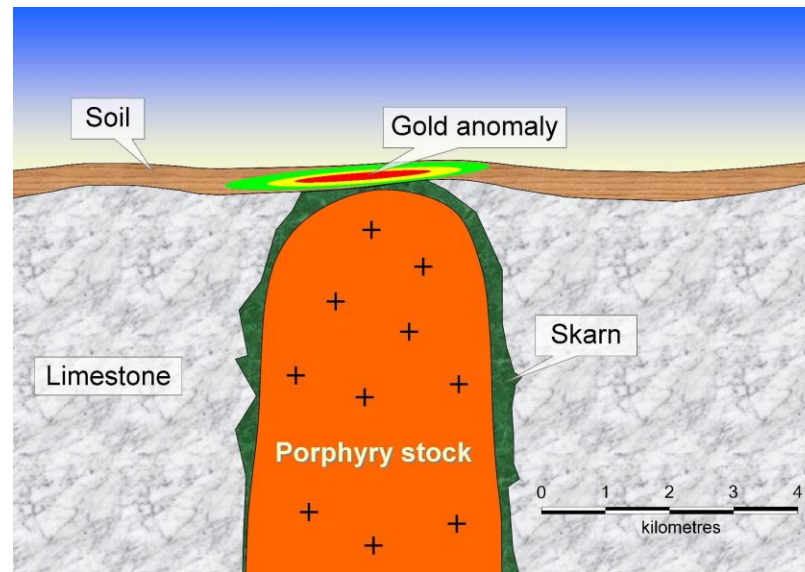
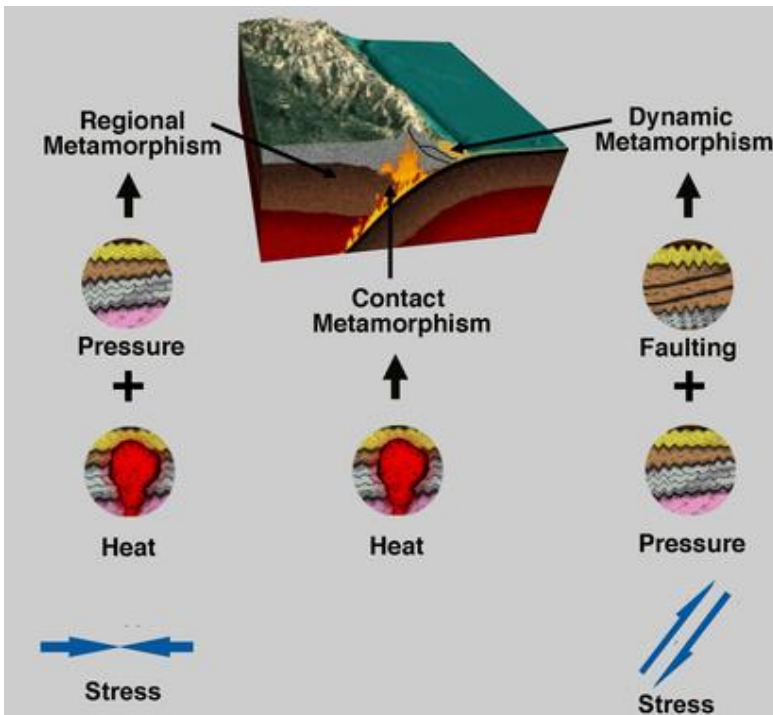
- When two mineral grains are squeezed together, the parts of their crystalline structures that touch are the most highly stressed.
- Atoms at these sites are readily dissolved by the hot fluids and move to the voids between individual grains.
- Thus, hot fluids aid in the recrystallization of mineral grains by dissolving material from regions of high stress and then precipitating (depositing) this material in areas of low stress. As a result, minerals tend to recrystallize and grow longer in a direction perpendicular to compressional stresses.





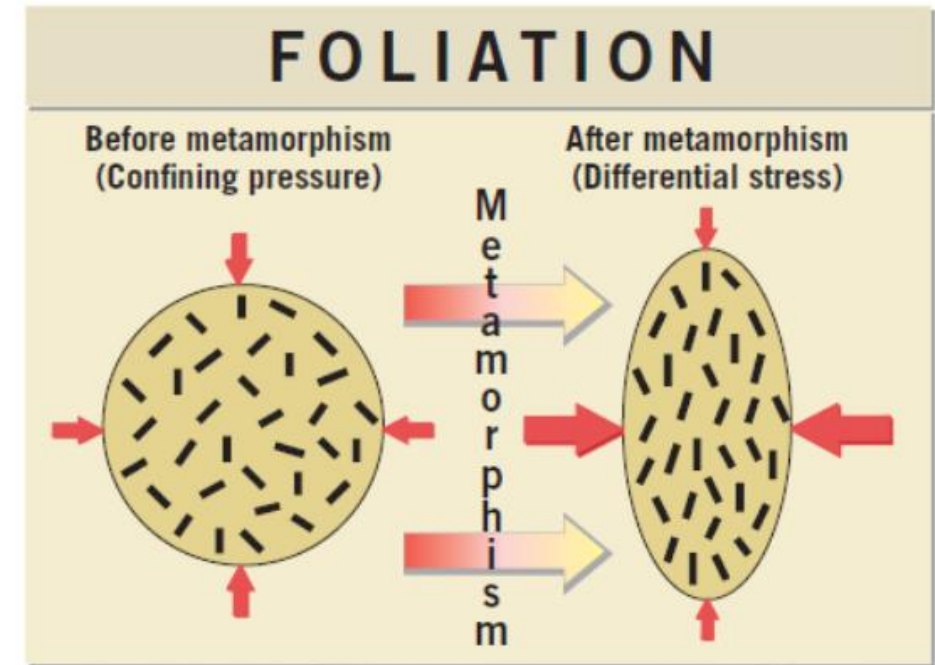
- When hot fluids circulate freely through rocks, ionic exchange may occur between adjacent rock layers, or ions may migrate great distances before they are finally deposited.
- The latter situation is particularly common when we consider hot fluids that escape during the crystallization of an intrusive mass of magma.
- If the rocks surrounding the magma differ markedly in composition from the invading fluids, there may be a substantial exchange of ions between the fluids and host rocks.
- When this occurs, the overall composition of the surrounding rock changes.

Types of Metamorphism



Metamorphic Textures

- The **degree of metamorphism is reflected in a rock's texture and mineralogy**. (Recall that the term texture is used to describe the size, shape, and arrangement of grains within a rock.)
- When rocks are subjected to **low-grade metamorphism, they become more compact and thus denser**. A common example is the metamorphic rock **slate**, which forms when **shale is subjected to temperatures and pressures** only slightly greater than those associated with the compaction that lithifies sediment.
- In this case, **differential stress causes the microscopic clay minerals in shale to align into the more compact arrangement found in slate**.
- Under more extreme conditions, stress causes certain minerals to recrystallize. In general, recrystallization encourages the growth of larger crystals. Consequently, many metamorphic rocks consist of visible crystals, much like coarse-grained igneous rocks.
- **Foliation refers to any nearly flat arrangement of mineral grains or structural features within a rock**.
- In metamorphic environments, **foliation is ultimately driven by compressional stresses** that shorten rock units, causing mineral grains in preexisting rocks to develop parallel, or nearly parallel, alignments.
- Examples of foliation include **the parallel alignment of platy** (flat and disk-like) minerals such as the micas; compositional banding, in which dark and light minerals separate generating a layered appearance; and rock cleavage, in which rocks can be easily split into tabular slabs.



Platy and elongated mineral grains having random orientation.



When differential stress causes rocks to flatten, the mineral grains rotate and align roughly perpendicular to the direction of maximum differential stress.

COMMON METAMORPHIC ROCKS

Metamorphic Rock		Texture	Comments	Parent Rock
Slate	Foliated		Composed of tiny chlorite and mica flakes, breaks in flat slabs called slaty cleavage, smooth dull surfaces	Shale, mudstone, or siltstone
Phyllite			Fine-grained, glossy sheen, breaks along wavy surfaces	Shale, mudstone, or siltstone
Schist			Medium- to coarse-grained, scaly foliation, micas dominate	Shale, mudstone, or siltstone
Gneiss			Coarse-grained, compositional banding due to segregation of light and dark colored minerals	Shale, granite, or volcanic rocks
Marble	Nonfoliated		Medium- to coarse-grained, relatively soft (3 on the Mohs scale), interlocking calcite or dolomite grains	Limestone, dolostone
Quartzite			Medium- to coarse-grained, very hard, massive, fused quartz grains	Quartz sandstone

- **Nonfoliated Textures:** Not all metamorphic rocks exhibit a foliated texture.
- Those that do not are referred to as nonfoliated and typically develop in environments where deformation is minimal and the parent rocks are composed of minerals that have a relatively simple chemical composition, such as quartz or calcite.
- For example, when a fine grained limestone (made of calcite) is metamorphosed by the intrusion of a hot magma body (contact metamorphism), the small calcite grains recrystallize and form larger interlocking crystals.
- The resulting rock, marble, exhibits large, equidimensional grains that are randomly oriented, similar to those in a coarse-grained igneous rock.

